



Water Content Evolution in the EDZ of Opalinus Clay: A Methodic Approach for a Comparative Interpretation of Measurements and Modelling

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Abstract

In the Mont Terri Rock Laboratory (Switzerland), an interdisciplinary examination program is carried out to increase knowledge about coupled hydro-mechanical effects in Opalinus Clay, which are of significant interest regarding the stability and integrity of a potential storage facility for high-level radioactive waste. This article focuses on the characterization of the claystone in the near field of excavations and related hydraulic effects due to excavation and ventilation. Electrical resistivity tomography (ERT) is applied to characterize the OPA: Several open fractures correlate with regions of high resistivity values, indicating potential preferential flow paths that are relevant for transport processes. Due to the combined interpretation of ERT long-term monitoring and seasonally repeated nuclear magnetic resonance (NMR) measurements, a relationship between electrical resistivity and water content can be established, resulting also in a time-dependent map of the water content around excavations with different climatic conditions. The statistical interpretation of these measurements indicates the existence of small-scale singularities in contrast to dominating, more homogeneous zones. The presented approach leads to a better process understanding of these heterogeneous near field effects and provides a valuable basis for a pragmatic approach to safety assessment.

Highlights

- Methodic approach to provide a time-dependent map of the water content around an excavation in the sandy facies of Opalinus Clay (OPA).
- Identification and statistical evaluation of potential flow paths (micro-cracking areas) in the near field of excavations with regard to numerical safety analysis.
- Establishment of the relationship between electrical resistivity and water content in the sandy facies of OPA based on combined evaluation of ERT and NMR measurements.
- Impact of heterogeneity and technical boundary conditions on the evolution of the water content and the permeability.
- Combined interpretation of multi-disciplinary methods aiming on a detailed process understanding and characterization of coupled hydraulic-mechanical effects in Mont Terri OPA.

Keywords Mont Terri project · Opalinus claystone · CD-A experiment · Coupled hydro-mechanical effects · Water content evolution · Geophysical measurements · Numerical modelling (OpenGeoSys 6)

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1 Introduction

Comprehension of coupled hydro-mechanical (HM) effects in claystone is of significant importance to evaluate the stability and integrity of the host rock of a potential deep geological repository for high-level radioactive waste (HLW). Especially in the near field of excavations, various effects like desaturation, swelling and shrinkage, plastic effects, and fracturing interact, which can lead to changes in porosity, permeability and retention properties.

This publication focuses on the combined interpretation of electrical resistivity tomography (ERT), nuclear magnetic resonance (NMR) measurements, the lithologic characterization and numerical modelling with regard to the evolution of the water content and micro-cracks as well as their impact on long-term effects on the safety in a repository. Further development of geophysical detection concepts, the set-up of related modelling approaches, and the definition of pragmatic methods for taking these effects into account in safety demonstration are of specific interest for the radioactive waste management (RWM) community.

Coupled HM effects in Opalinus Clay are induced and affected by excavation as well as the ventilation conditions. As desaturation processes significantly affect the deformation behavior, this paper focuses on hydraulic effects and aims on providing a comprehensive basis for further development of numerical modelling approaches in the near field of excavations. These effects have also been investigated and discussed in Maßmann (2009) for the claystone in Tournemire, in Peron et al. (2010), Bond et al. (2013), Mayor et al. (2007), and Williams et al. (2022) related to investigations in the Mont Terri Rock Laboratory and in Navarro (2005). The presented work is carried out in the context of the Cyclic Deformation (CD-A) experiment (Influence of humidity on cyclic and long-term deformations) Ziefle et al. (2017) as part of the Mont Terri Project Ziefle et al. (2021).

The aim of the CD-A experiment in general is the evaluation of the hydraulic influence on the long-term deformation behaviour of claystone. The unique setting of two identical constructed twin niches, differing only in their ventilation, enables the investigation of the near field effects like the desaturation and the evolution of the EDZ. In this context, the impact of the lithology on the subfacies scale on intrinsic properties like resistivity on one side and process-variables like water content on the other side are addressed (see also Amabile et al. (2020)). The research aims at a detailed, time-dependent description of the extent and hydraulic characteristics of the EDZ with regard to potential flow paths. The interdisciplinary interpretation is carried out in view of the creation of numerical models, which are to be set-up, calibrated and verified by means of the various investigation (experimental) methods. Besides the presented work,

focusing on water content evolution in the near field, the CD-A experiment in general is accompanied by a comprehensive measuring program including geological, geotechnical and geophysical characterization as well as numerical modelling. The measuring program includes measurements of the permittivity (Taupe-TDR), the permeability (Pulse tests), different types of water content measurements (NMR, CCM), suction measurements (Thermo-hygrometer and psychrometer), pore pressure measurements (Mini-Piezometer) and deformation measurements (extensometer, laser scans) up to a depth of about 4 m (Ziefle et al. 2024). It focuses on an increased process as well as system understanding of coupled HM effects in OPA.

In the literature, many publications related to the characterization and process understanding of coupled effects in claystone can be found (e.g., Wild and Amann 2018; Wild et al. 2017). Numerical procedures have been developed, implemented, tested and validated against experiments and will be a valuable tool for subsequent safety assessment cases which are still under development (please refer to Armand et al. (2017), Seyedi et al. (2021) amongst others). Manifold geophysical investigations have been performed in the laboratory as well as in-situ focusing on the lithologic description of the claystone, the material parametrization of claystone and EDZ and furthermore the evolution of material parameters, water content and crack evolution. From the geophysical point of view, seismic measurements are a widely used and acknowledged tool for detailed structural and mechanical in-situ characterization. Esefelder et al. (2021) demonstrates the seismic anisotropy of the shaly and carbonate-rich sandy facies of Opalinus Clay in the Mont Terri Rock Laboratory at a depth of 2 m away from excavations, suppressing effects caused by the excavation damaged zone. Williams et al. (2022) investigates the evolution of the EDZ and self-sealing for excavations of different age with seismic measurements, indicating sealing of the host rock over decades. Further development of laboratory and in-situ testing procedures for the investigation of material properties related to coupled HM effects is ongoing and can also be found in Amann et al. (2017, 2018), Bock (2009), Zhang and Laurich (2020), Zhang et al. (2022), Chen et al. (2022) and Jiang et al. (2022). Additionally, resistivity measurements are used to study on the one hand the microstructure of the material and on the other hand changes in the fluid content (Tang et al. 2022). Crisci et al. (2021) proposes a correlation of the HM response to the mineralogical variability of the tested specimens, defined as shaly (high clay-mineral content) and sandy (low clay-mineral content) layers. On the laboratory scale it is shown that by adopting a layered structure with an alternation of shaly and sandy layers, most of the variability in the studied geomechanical properties of Opalinus Clay can be captured. Similar approaches may also be used on a larger scale. The work at hand contributes

in an increased understanding of the in situ water content evolution and related effects on the barrier integrity and can be seen as basis for the parametrisation of permeability and tendency to cracking. Extensive studies on this exist with regard to the effects in the Callovo-Oxfordian (COx) claystone which is investigated in the French radioactive waste disposal project. Alcolea et al. (2017) amongst others discusses possible characterization strategies and Armand et al. (2013) contributes a comprehensive study of the initial situation around various drifts in the COx.

The significant impact of the water content on the mechanical properties like stiffness and strength has been highlighted by Ip and Borja (2022): For transversely isotropic rocks, the effects of saturation can differ between the bedding directions, which gives rise to a saturation-dependent stiffness and strength anisotropy, often with larger stiffness and strength after drying. The accurate prediction of mechanical behaviour of clay rocks under partially saturated conditions requires numerical models that can capture the evolving elastic and plastic anisotropy depending on the water content. Through numerical simulations with an anisotropic modified Cam-Clay (MCC) model, Ip and Borja (2022) demonstrates the role of evolving stiffness and strength anisotropy and the impact of the saturation on the mechanical behaviour of clay rocks. The response of OPA to relative humidity changes has been investigated by Wild et al. (2017), indicating irreversible expansion normal to bedding. The strength and water retention characteristic remain unaffected by the relative humidity variations.

Related to the geophysical application, the investigation of lithology but also the fractured zones is of specific importance as these zones may come along with a change of material parameters affecting the hydraulic and mechanical processes.

In general, many aspects of hydromechanical behaviour of claystone are under discussion with respect to experimental analysis, mathematical description and numerical treatment. Numerous international projects and publications deal with this topic as can be seen in Armand et al. (2017), Plúa et al. (2021), Bond et al. (2013) and Kolditz et al. (2021) amongst others.

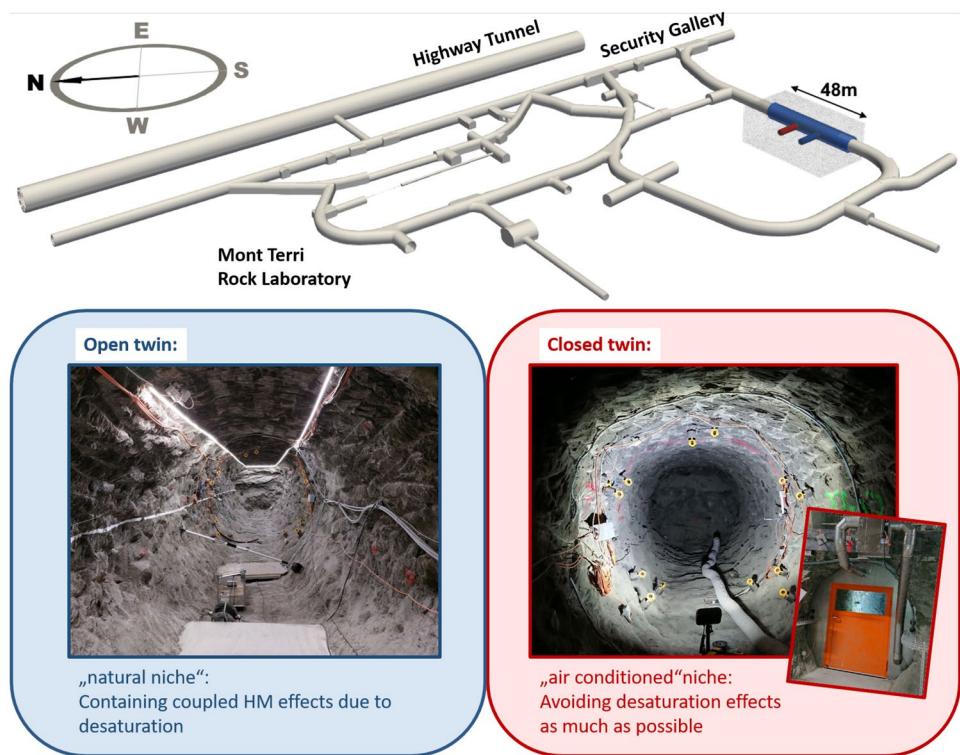
This article summarizes results of the CD-A experiment during the first two years after excavation aiming on a comprehensive interpretation of the observed water content evolution of the near field. In general, laboratory investigation results depend on the preparation procedure and extraction location of the core samples. The methodic approach presented here gives (a) an overview of the distribution of initial heterogeneities in the near field, (b) provides a moisture map, giving information about the distribution of the water content over the time and (c) interprets the water content measurements statistically,

indicating a heterogeneous evolution of the water content in the near field. The interpretation of this results is carried out without distinguishing between EDZ and undisturbed host rock. The approach enables the interpretation of water content changes with regard to the lithologic description of the rock surface and the seasonal change of humidity conditions. The novel opportunity of combining long-term 2D ERT with point-wise water content measurements by nuclear magnetic resonance (NMR) enables the option of transferring the resulting electrical resistivity maps into maps of water content. The corresponding transfer function has a reliable, i.e., physical, background and is directly driven by in-situ measurements. As demonstrated in this paper, such data sets can serve as a basis for testing, correcting, and permanently adapting the numerical modelling approaches including their boundary conditions.

2 CD-A Experiment in the Mont Terri Project

The Mont Terri Project is an international research project aiming at the geological and geotechnical characterization of Opalinus claystone in a rock laboratory. It is located in the western part of Switzerland in the Jura mountains and hosts various field experiments to investigate the coupled thermal-hydro-mechanical and chemical processes in the Opalinus Clay. The results provide general insights about the suitability of clayrock as a potential host rock for radioactive waste disposal or other geotechnical applications like Carbon Capture and Storage (CCS). More information about the geology and lithography of the Opalinus Clay in the area of the Mont Terri Rock Laboratory can be found in Bossart et al. (2018), Hostettler et al. (2018), Nussbaum et al. (2018) and Jaeggi et al. (2018). The geomechanical behaviour at multiple scales is summarized in Amann et al. (2018). Manifold information about the Mont Terri Rock Project is also given in Bossart et al. (2018). The excavation of the CD-A twin niches has been carried out as part of the extension of the Mont Terri Rock Laboratory (Bossart and Burrus 2019) in September 2019. The niches were excavated in fall 2019 near the Gallery 2018 in an area characterized by the upper sandy facies of the Opalinus Clay as shown in Fig. 1. With a diameter of 2.3 m and a length of 11 m, both are oriented perpendicular to the strike of bedding. While one niche (open twin-color coded in blue) is ventilated in the same way as the rest of the underground laboratory, the other niche (closed twin-color coded in red) is sealed off so that a uniformly high humidity is maintained. The surface of the claystone in the niches is not covered with any kind of stabilizing support. Consequently, a unattenuated interaction between the climatic conditions in the niches and the claystone is to be expected, the possible consequences of which should

Fig. 1 Top: Location of the CD-A experiment in the Mont Terri URL (top) and view into the open (bottom left/blue frame) and the closed (bottom right/red frame) niche



be directly observable and even measurable. This model setup allows the study of the effects of climatic conditions on the behaviour of the host rock. The experiment is characterized by a comprehensive measuring campaign which is combined with numerical modelling and the use of visualisation techniques.

The complex measuring program of the experiment focuses on the climatic conditions within the twin niches, the characterization of the host rock, and the evolution of the deformation, saturation and pore pressure as it is presented in Ziefle et al. (2019) and Maßmann et al. (2019) and discussed in Ziefle et al. (2024). Within a few weeks after excavation, a complex measuring system in and around the niches has been installed and the climatization has been adjusted in the closed twin as described in Ziefle et al. (2021). A comparative interpretation of the investigated parameters provide information concerning the distribution of different litho–stratigraphic (sub)units, the extent of a excavation/borehole damaged zone (EDZ/BDZ) as well as the occurrence and properties of tectonic fault zones and individual faults and is discussed in Ziefle et al. (2024).

The temperature, atmospheric pressure and relative humidity of the air within the two niches and the adjoining gallery are measured permanently at several locations (Schwab et al. 2021). In each niche, 5 measuring devices (Relative humidity and temperature EE210 by E + E Elektronik GmbH) at different locations are installed. They have a measuring

range for humidity between 0 and 100% with an accuracy of $\pm (1.3 + 0.003)\%$ RH. The measured climatic conditions indicate nearly homogeneous conditions within each niche, with slightly higher relative humidities at the bottom and the back part of each niche. They are presented for both twin niches in Fig. 2. The open niche is characterized by seasonal evolution of temperature and relative humidity with higher values for these quantities in summer and lower values in the winter season. In contrast, the climatic conditions in the closed niche tend to a constantly high relative humidity of nearly 100% and equilibrium mean temperature of about 13.8 °C. Furthermore, the measurements indicate clearly the impact of calibration of air condition within the first months (2019) and of opening times of the sealing door. As intended, no seasonal variation on temperature and relative humidity can be identified in the closed niche. The temporal evolution in both niches indicates an impact of excavation at the early measurements. Here, an initially higher temperature can be identified.

Additional measurements focusing on the deformation behaviour such as laser scans, convergence and extensometer measurements are not part of this article but are discussed in combination with geologic interpretations, numerical modelling approaches and visualization techniques in sigeom (2020), and Ziefle et al. (2022). Additional measurements of crack evolution have been performed using scanline measurements and summarized in Regard et al. (2022). Investigations focusing on the pore pressure evolution in the saturated host rock conducted by mini-piezometer, as well as suction

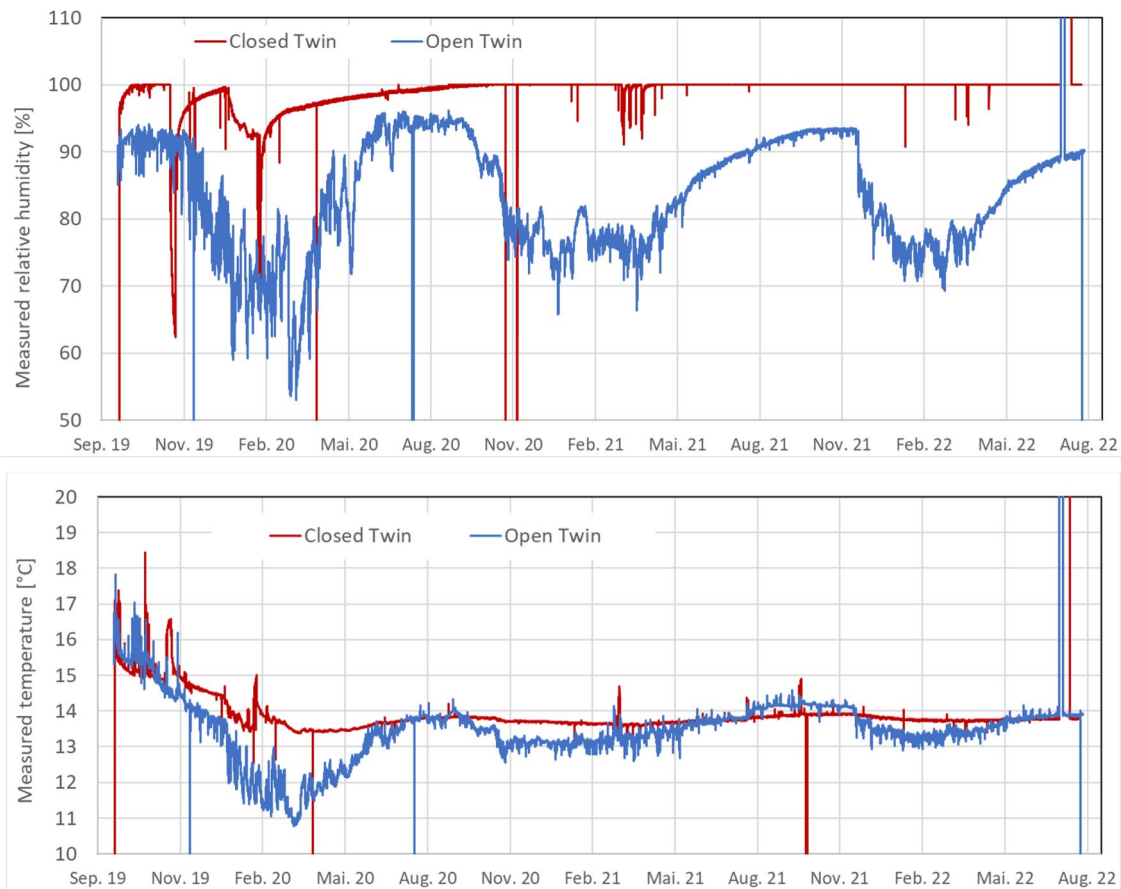


Fig. 2 Measured relative humidity (RH) (top) and temperature (bottom) in the twin niches. Lower RH within the first couple of months results from installation periods as well as technical issues with the ventilation of the niche

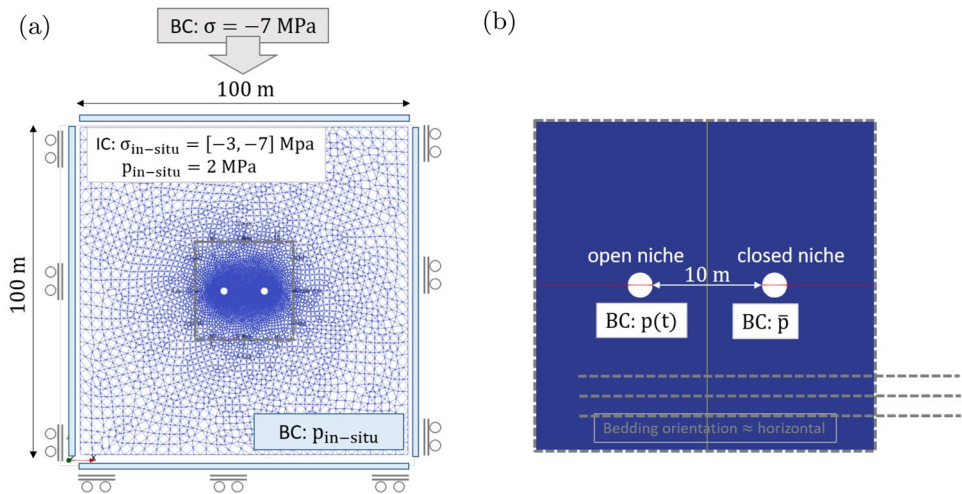
measurements in the desaturated area conducted by psychrometers and thermo-hygrometers are in preparation. The interpretation of this measurements in combination with numerical modelling procedures also focuses on the improvement of constitutive models e.g., the retention curve. In this context, seasonal effects related to saturation and desaturation of the rock and its influence on the deformation behaviour, stability, host rock integrity and consequently its containment capability need to be considered. Swelling and shrinkage, elasto-viscoplastic material behaviour as well as damage and fracturing should be taken into account.

3 Numerical Modelling Approach of Near Field Hydraulic Effects

A numerical model is applied to investigate the behaviour of the rock at the niches. The open-source scientific software OpenGeoSys (OGS) (Kolditz et al. 2012, 2021; Bilke et al. 2022) has been used to setup a HM coupled two-dimensional model. The model is based on an unsaturated HM

approach. Since the focus of the current work is placed on the desaturation effects, we do not discuss the mechanical/hydro-mechanical effects and refer the reader to additional publications, e.g., comparison of monitored and modeled convergence Zieffle et al. (2022), desaturation and strain evolution Cajuhi et al. (2023a). The chosen macroscopic HM approach is based on poro-elasticity with the momentum balance of the solid and the mass balance of the pore fluid (Gawin and Schrefler 1996; Zieffle et al. 2017; Kolditz et al. 2021; Pitz et al. 2022). The coupling between the solid and liquid parts is performed with the effective stress concept (BIOT) and the Bishop's approximation (Bishop and Blight 1963; Gawin and Schrefler 1996; Lewis and Schrefler 1998). Linear elasticity and modified Darcy flow are applied, taken account desaturation by the Richards equation (Richards 1931). Following Richards, the pore gas pressure is kept constant, while the liquid pressure can vary. As the pore gas pressure is assumed to be zero, negative pore pressures equal capillary pressures. Using the van Genuchten curve (Van Genuchten 1980), the liquid saturation S_{eff} can be defined as a function of capillary pressure p_{cap} :

Fig. 3 Numerical model setup: whole domain (a) and zoomed view (rectangular area) of the niches (b). The open niche is subjected to seasonal boundary conditions, while the pore pressure at the closed niche is approximately constant. No displacement boundary conditions as well as a constant, in-situ measured, pore pressure are applied around the boundaries of the two-dimensional model



$$S_{\text{eff}} = \left(1 + \left(\frac{p_{\text{cap}}}{p_b} \right)^n \right)^{\frac{1}{n}-1}, \quad (1)$$

with $p_b = 10$ MPa, $n = 1.82$ and the residual saturation $S_r = 1$.

In the numerical model setup the twin niches are placed symmetrically in a two-dimensional domain with $100 \times 100 \text{ m}^2$ with 8510 quadratically interpolated Finite Elements as depicted in Fig. 3. The material properties utilized in the simulation correspond to the sandy facies with anisotropic hydraulic and mechanical behaviours obtained from the literature (Bossart et al. 2018,[51]). The anisotropy of the intrinsic permeability is modeled by a factor of 2.5 as derived from the hydraulic conductivity given by [51]. Initially, the stress-state as measured in-situ (Martin and Lanyon 2003; Ziefle et al. 2017) is projected on the mesh with the compressive total stresses $\sigma_{\text{horizontal}} = 3$ MPa and $\sigma_{\text{vertical}} = 7$ MPa. The initial liquid pore pressure is set to 2 MPa according to Garitte et al. (2017) over the domain, which also corresponds to the outer boundary condition. Moreover, no displacement normal to the edges of the outer domain are allowed by boundary condition. No further mechanical loading is assigned to the model, consequently, the main driving load is the hydraulic part. In order to define the transient hydraulic boundary conditions around the open niche, the RH depicted in Fig. 2 is used. The Kelvin equation Thomson (1872) is applied to determine the capillary pressure based on measured RH and temperature. Since one of the primary variables of the model is the liquid pore pressure, the Kelvin equation (given in Thomson (1872)) is used to transform relative air humidity into pore pressure to be used as boundary condition in the open twin:

$$p = \frac{\rho_L RT}{M_L} \ln(\text{RH}) \quad (2)$$

The liquid density, the universal gas constant and molar mass of the pore fluid are given by $\rho_L = 1000 \text{ kg/m}^3$, $R = 8.314 \text{ J/(mol K)}$ and $M_L = 0.01802 \text{ kg/mol}$, respectively.

However, one crucial point in modelling the desaturation effects around the twin niches is the behavior at the surface of the host rock. Here, various effects come into play and interact. The physical and mathematical description, which are basis for the numerical modelling, are complex and the related interaction of effects is not yet fully understood. The mentioned approach is the common state-of-the-art approach and has to be critically questioned. The RH variations in the closed niche observed in the first months of the experiment are related to door openings, e.g., installation of measurement sensors. Nevertheless, the pore pressure is set constant and is calculated with the average values of the maximum monitored RH as it reflects the situation since May 2020.

As a result of the numerical calculations, the relative saturation changes near the open niche are depicted in Fig. 4 for September 2020 and March 2021. The temporal evolution of these changes are presented in Cajuhi et al. (2021b). A

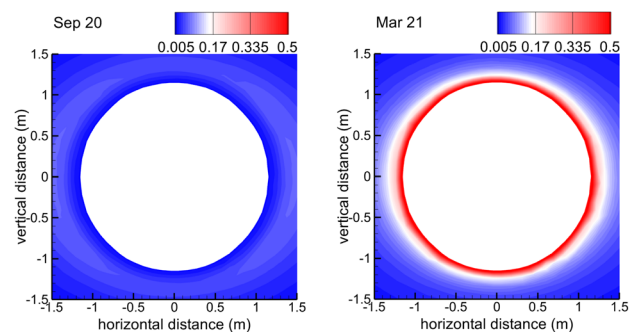


Fig. 4 The difference between the saturation values with respect to the reference state in October 2019 are computed for the open twin. The results for September 2020 (a) and March 2021 (b) are shown in analogy to Fig. 5

decrease of saturation is predicted for the winter period. The saturation is lowest at the end of winter, which is in accordance to our observations from the geophysical results (please refer to Sect. 5). The calculated saturation changes at the surface and in the near field of the open niche come along with a water vapour inflow from the rock into the niche. The amount of this water vapour flow can be recalculated from the model and lies in the range of 0.5 to -2.5 l/day; depending on the season and on the time since excavation. In comparison to the values from the open twin, the water vapour flow from the closed twin only comprises a decrease of the water vapour flow depending on the time since excavation and lies in a range of about 0.2 l/day. This flow rate is related to initial effects. Since the pore pressure applied in the closed niche is constant, no changes in saturation are calculated and the results of the closed niche are not plotted. This kind of investigations is especially interesting for the long term evolution around the niches and is a still ongoing work which is accompanied by in situ measurements focusing on the tightness of the sealing door in the closed twin.

Since the model is homogeneous, the computed saturation values S_w are constant circularly around the niche. These results are used to compute the monthly volumetric water content θ [m^3/m^3] with

$$\theta = nS_w, \quad (3)$$

where $n = 0.105$ [–] is the water loss porosity estimated in Bossart et al. (2018). In summary, the presented homogeneous model approach enables the investigation of desaturation effects due to climatic conditions. In the following, the impact of heterogeneous effects and numerical model calibration are discussed. We focus on a combined interpretation of monitored data and modeling approach.

4 In-Situ Investigation Program

Important process quantities regarding coupled HM effects in claystone are displacements, stress field and pore pressure, amongst others. These parameters are part of the CD-A measuring program as discussed in Zieflé et al. (2022) and Maßmann et al. (2019). However, the water content within the host rock is a key parameter for various processes affecting the mechanical behaviour. As the work at hand focuses on the effect of desaturation near the excavation, the presented in-situ investigations aim on the one hand on a characterization of the host rock as also discussed in Zieflé et al. (2024), and on the other hand on the evolution of the water content around the twin niches. The characterization of the host rock is carried out with the electrical resistivity tomography (ERT) in a range of decimeters around the niches and by a geological interpretation at the surface along the niche

profile. Furthermore, the relative change of electrical resistivity provides information about the water content changes around the niches. Additional measurements of the water content evolution are carried out by nuclear magnetic resonance (NMR) measurements investigating the water content evolution near the surface.

4.1 Characterization of the Host Rock by ERT

ERT is used to provide information about the material composition and the change of water content. For that purpose, a vertical ring consisting of 72 equidistant electrodes on the wall in each niche was installed in October 2019. The resistivity measurements along this ring profile are repeated daily and enable an interpretation of structures larger than the distance between the electrodes. Smaller scale effects may be measurement-related artefacts.

The setup contains all possible Wenner- α and Wenner- β configurations, respectively (Knödel et al. 2007). This allows observing the electrical resistivity and its changes up to a depth of a few decimeters. A numerical inversion process is necessary for data analysis and interpretation. The measured apparent resistivities at the niche wall have to be transformed into a spatial model, discretised into a distinct number of elements of homogeneous resistivity, taking into account the electrode geometry. The forward operator in the inversion process is generally obtained by finite-difference (FD) or finite-element (FE) methods. For our data, the non-commercial software package BERT (Boundless Electrical Resistivity Tomography) is used.¹ For further details, see Günther et al. (2006) and Rücker et al. (2006). Windows of $3 \times 3 \text{ m}^2$ are chosen to depict the ERT results to focus on the region near the surface, where the resolution of the method is optimized given the setup described above.

The measured reference electrical resistivities (ERT) (measured at 31.10.2019) are presented in Fig. 5 (left) and indicate the heterogeneous material composition of the sandy facies of the OPA around the twin niches. Here, areas with higher carbonate content are characterized by higher electrical resistivities. The temporal evolution of this measurement is depicted in Fig. 5 (middle and right). Based on the measurements at the reference day, the relative changes of the resistivity distributions are exemplarily plotted and compared for two days in August 2021 and 2022, indicating zones with high changes of the measured resistivity, especially with a significant increase of the resistivity (red zones). This increase of resistivity is directly related in time to seasonal effects and can therefore be interpreted in two ways: On the one hand, the measured higher resistivities can result directly from a lower water content. On the other hand,

¹ <https://gitlab.com/resistivity-net/bert>.

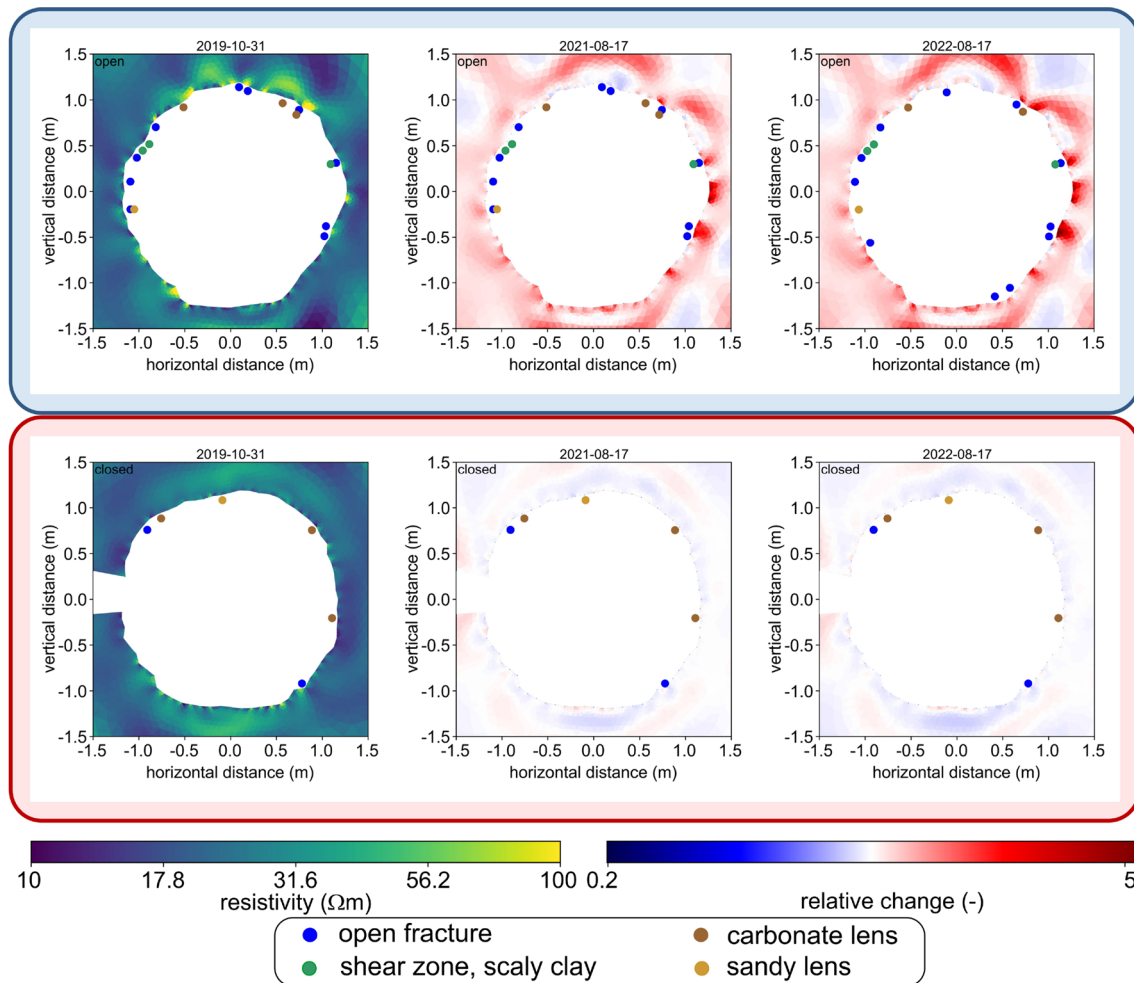


Fig. 5 Reference electrical resistivity (ERT) (measured at 31.10.2019) along the open (top left) and closed (bottom left) niche profile and relative change [–] of the resistivity related to the reference measurement for the open (top) and the closed (bottom) twin

measured in August 2021 (middle) and August 2022 (right). The results of the geologic interpretation (carried out by swisstopo) are marked by the colored dots along the niche profiles

a seasonal geometric change in the fracture network—potentially coming along with corresponding changes in hydraulic properties such as permeability—is also conceivable. Whatever the case, the fact is that the areas are characterised by a seasonality coming along with a varying ratio between solid rock and fluid. Finally, they are interpreted as changes in the water content, even if they potentially come along with seasonal geometric changes of the fracture network. Additionally, significant differences in the desaturation process around the twin niches become obvious. In the open niche, changes of the resistivity that potentially correspond with a desaturated zone with an extent of some decimeters can be observed, which is influenced by the seasonal variations of the climatic conditions described above. In contrast, the changes of resistivity (and consequently water content) are

very low in the closed niche. In general, the results in the open twin also indicate stronger desaturation after the cold and dry winter months than after the warmer and wetter summer months. This fact becomes obvious in the statistical interpretation discussed in Sect. 5. With regard to the geometric distribution of positions with high resistivities and high relative changes, the arrangement follows the direction of the strike of bedding and indicates a strike direction of about 145° corresponding to the literature (Ziefle et al. 2022). For all electrode positions of the ERT ring profile the rock has been analyzed regarding the existence of carbonate or sand lenses, shear zones or scaly clay as well as open fractures. The results are plotted by the coloured dots in Fig. 5. The general observation of higher electrical resistivity in areas with higher carbonate content is already documented

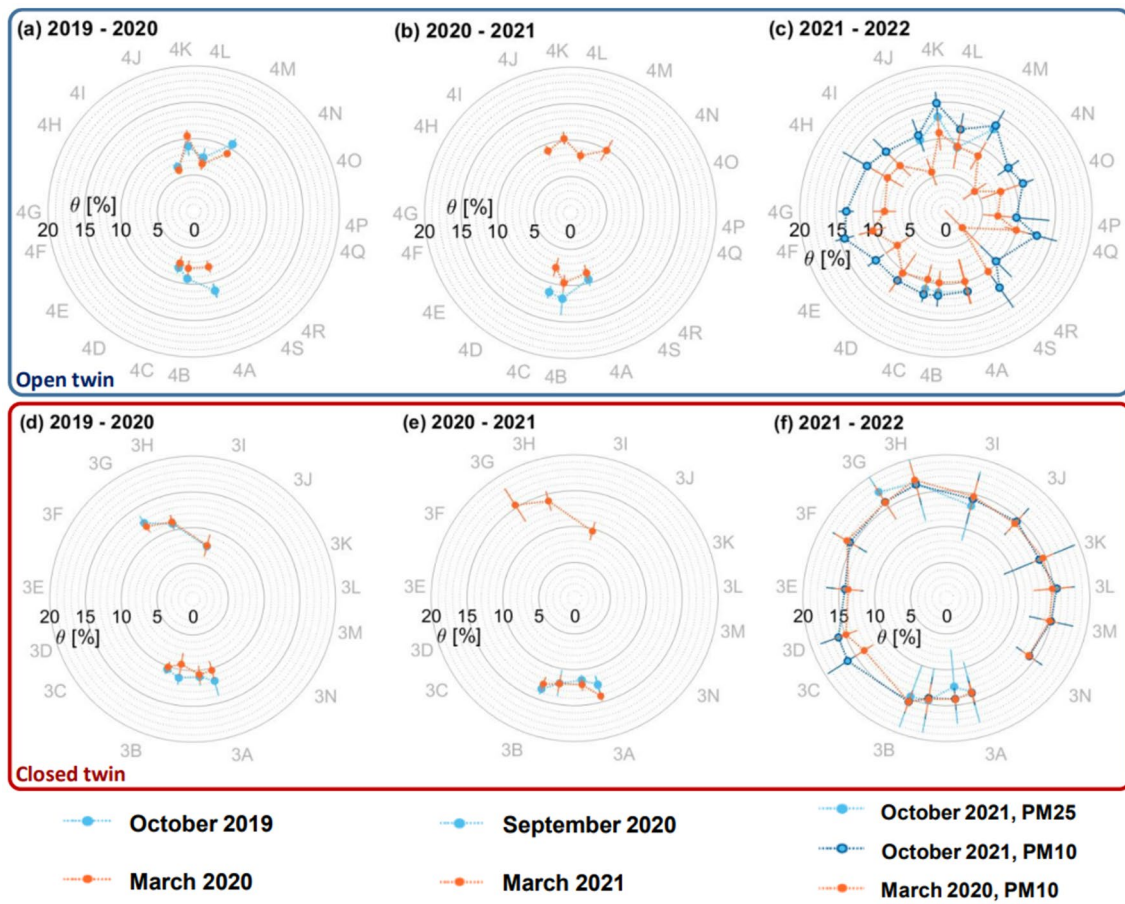


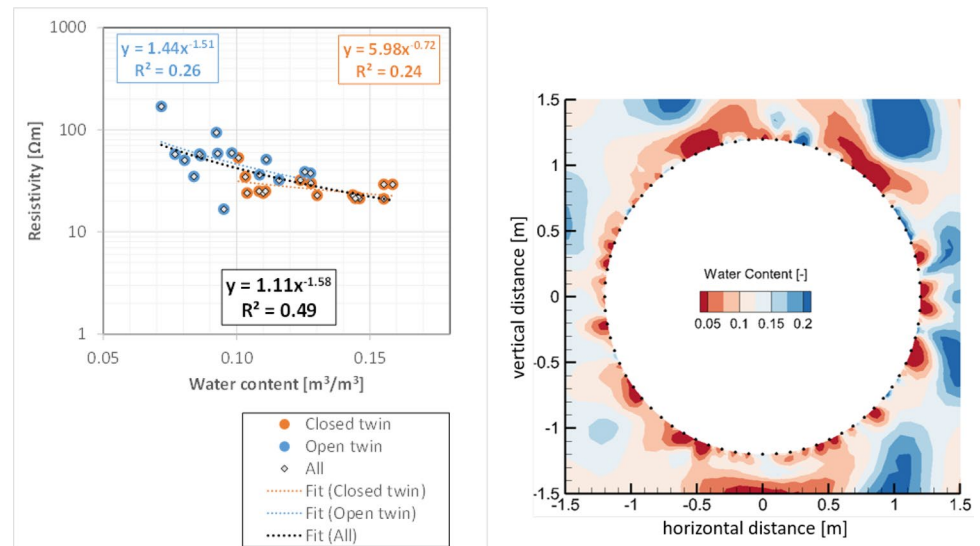
Fig. 6 The volumetric water content as measured about every six months using non-invasive single-sided NMR is depicted as radii in these circle diagrams, which schematically indicate the niche cross sections. The measurement points 4A to 4S (open twin, top) and 3A to 3N (closed twin, bottom) twin appear at the angular values corresponding to the real positions inside the niches (Ziefle et al. 2021).

in Kneuker et al. (2017) and also discussed in Ziefle et al. (2024). While an impact of carbonate/sandy lenses and shear zones/scaly clay on resistivity outliers can not be identified, some of the detected positions are characterized by open fractures and additionally by high initial resistivities (measured in October 2019), often corresponding with a strong increase of resistivity over time. The observed correlation between open fractures and resistivity is based on very rare data points, but indicate that 70% (7 out of 10) of the detected locations with existing open fractures correspond to areas with high initial resistivities suggesting areas with high carbonate content and a (potentially resulting) significant increase of the resistivity over time. Nevertheless, also open fractures with lower resistivities as well as unfractured regions with higher resistivities are found in the data set. The

From 2019 to 2021, the NMR-Mouse PM25 (Magritek) was used for sampling a depth of approximately 1 cm on floor and ceiling, whereas since October 2021 the NMR-Mouse PM10 (Magritek) has been utilized that samples a depth of 0.6 cm all along the profile of the twin niches. Please see the text for further details

seasonal variation of the amplitude of these high resistivities can be associated to changes in crack properties, like crack aperture, which is reported to occur due to desaturation (see Maßmann 2009; Ziefle et al. 2017). Additional scanline measurements, which are also carried out in the CD-A experiment, give information about the evolution of aperture and length of desiccation cracks at the surface of the host rock. These investigations also indicate a seasonal variation of the mentioned crack properties. Nevertheless, this is just an observation of the surface, while the measurements discussed in this paper focus on the decimeter scale around the niches. Nevertheless, these results are a motivation to use additional numerical approaches like phase-field models (Cajuhí et al. 2018, 2023b; Heider and Sun 2020) in the modelling approach.

Fig. 7 Crossplot of electrical resistivity (ERT) and volumetric water content (NMR) as measured at the surface of the twin niches (left). Water content distribution around the open twin as interpreted from ERT measurements in April 2021 (right) based on the power law relation of the generalized Archie model according to Shah and Singh (2005)



4.2 Water Content Evolution by ERT and NMR Monitoring

As mentioned before, the permanent ERT installation has provided a complete measurement every day since October 2019. The relative changes of the measured electrical resistivity are of main interest, because these are directly associated with saturation/desaturation processes in the OPA adjacent to the niche. Additional water content measurements have been performed by the NMR method, which delivers a response signal of the ^1H nuclei in water molecules after electromagnetic excitation. If exposed to an alternating magnetic field oscillating at the Larmor frequency, the ^1H proton spins are forced to tilt away from their equilibrium positions in a static magnetic field, and relax back to it after shutting off the excitation field. This relaxation process is observed as exponentially decaying signal (Coates et al. 1999). The more ^1H protons are included in this process, the stronger is the detected signal—a simple linear relationship that can easily be transferred to water content estimates. We apply the single-sided NMR technology for non-invasive in-situ measurements directly on the wall of the CD-A niches. For details on this relatively new NMR measurement technique and its potential use in geosciences, we refer to (Blümich et al. 2008) and (Costabel et al. 2022). In principle, NMR is able to distinguish different kinds of water in rocks, i.e., adsorbed and free pore water, if the signals from the corresponding proton spins decay with different rates. Consequently, NMR signals from porous geo-materials are generally assumed to be the superposition of different exponential decays (Coates et al. 1999). When clay minerals are involved, even proton spins of hydroxyl groups being part of the (pseudo-) solid rock matrix might contribute to the measured NMR signal (Fleury and Romero-Sarmiento 2016). However, as demonstrated by Costabel (2023), the equipment used in the

CD-A experiment does not allow a robust multi-exponential analysis for NMR data from Opalinus Clay, whereas its mono-exponential representation is dominated by the pore water and leads to water content estimates being consistent with the amount of the water lost after drying at 105 °C. This is because the assumption of a single exponential component represents an averaged decay that suppresses the fast-decaying signal components from interlayer water and/or hydroxyl groups.

Using the single-sided NMR in combination with the air humidity measurements in the twin niches, we investigate the seasonal moisture interactions between niche atmosphere and rock surface. In addition, we anticipate that the in-situ NMR data from the niche surfaces support the quantitative interpretation of the ERT monitoring with respect to moisture variations inside the OPA directly adjacent to the niche wall.

From October 2019 to October 2021, we used a single-sided NMR device with a 40 by 40 mm² sensor for exciting and detecting an NMR signal with optimal signal-to-noise ratio (NMR-Mouse PM25, Magritek). The sensitive area, a footprint with the same dimensions as the sensor coil and a thickness of 200 μm , is placed outside the device. The apparatus was adjusted to sample the volumetric water content at a depth of 1 cm of the material in front of the sensor. Specific NMR measurement points were determined and

Table 1 Empirical values according to the generalized Archie model (Eq. 4) for the in-situ data set

Data specification	c	m	R^2
Open twin	5.98	0.72	0.24
Closed twin	1.44	1.51	0.26
Both niches	1.11	1.58	0.49

positioned in form of a ring around both niche walls (Zieffle et al. 2021). The distance of these points to the ERT profile differ between 20 and 50 cm. An equidistant determination of the NMR points with respect to the ERT profiles was not possible because we had to consider the roughness of the niche surface that does not allow an arbitrary placement of the NMR device. We successfully applied the device described above on the measurement positions of floor and ceiling. Due to the weight of the device (35 kg), the attempts to conduct measurements also on the positions of the side-walls were assessed to be too risky for health and material and were stopped for safety reasons. Since October 2021 a 6 kg lightweight version of a single-sided NMR device (NMR-Mouse PM10, Magritek) has been acquired that also allows measurements from the sidewalls. This advantage comes at the cost of a lower signal-to-noise ratio. The sensor of the lightweight device has a size of 16 by 16 mm² and is adjusted to sampling a depth of 0.6 cm.

To capture seasonal changes of the moisture content at the NMR measurement points, we acquire data at least two times per year (summer and winter). Figure 6 visualizes and compares the corresponding results for the twin niches since the beginning of the experiment. The circle diagrams indicate schematically the cross sections of the niches with the measurement points at the exact angular positions, while the volumetric water content is depicted as the radii of concentric circles. We observe that the results in the open twin indicate an overall drier wall than in the closed twin for the entire year. The water content in the open twin ranges from less than 10 vol% in March to more than 10 up to 15 vol% in October. In contrast and as expected, the water content at the wall of the closed twin hardly changes with season. However, we have observed a slight continuous increase since October 2019 with starting values between 8 and 13 vol% during the first year up to 13–16 vol% in 2022 here. This trend has halted in 2021 and the water content appears to have stabilized at each observed point.

4.3 Combined Evaluation of ERT and NMR

As described, NMR measurements are used to directly measure the water content at a very shallow depth of less than 1 cm on certain points around the niches while ERT measurements probe the electrical resistivity distribution up to depths of a few tens of cm. The following consideration focus on combining the two for extending the water content information. Pore water in porous rocks and the dissolved ions in it create flowpaths for electrical current, thus an increasing water content usually reduces the electrical resistivity. An adequate description of this relationship is given by the generalized Archie-model (Shah and Singh 2005), which links the two quantities by a relatively simple power law:

$$\rho = c \cdot \rho_{\text{fluid}} \cdot \theta^{-m}, \quad (4)$$

where ρ and ρ_{fluid} [Ωm] are the measured resistivity and that of the pore fluid, respectively, and θ is the water content [m^3/m^3]. The dimensionless parameters c and m are empirical variables, which, according to Shah and Singh (2005), can range within 0.3–15.9 and 0.7–3.9, respectively. If c and m are known or can be identified by sufficient calibration data, ρ -measurements can provide water content estimates.

We test this option using the in-situ ERT and the NMR measurements from the surfaces of the twin niches. Figure 7a shows a corresponding semi-logarithmic cross plot of θ and ρ . Because the positions of the ERT-electrodes and the NMR measurement points are not the same, a reasonable connection between the measurement results is not straightforward. Of course, we consider only the ERT results from those dates, on which the NMR measurements took place. However, the spatial connection is an issue. Considering the fact that the ERT data demonstrates significant differences in the temporal behaviour of different segments around the niches as presented in Fig. 5, we spatially average the ERT inversion results at the tunnel surface separately for ceiling, floor, left, and right sidewall for the considered dates. In doing so, we only take into account those parts of the cross-sectional arc, which are also covered by NMR measurement points. The NMR water content results depicted in Fig. 6 are averaged correspondingly.

Finally, this procedure leads to several $\rho - \theta$ data pairs, which can be fitted with equation (4) to test how well it describes our data. Regarding the coefficient of determination R^2 , between 0.2 and 0.3 individually for each niche and 0.5 for the whole data set, we can conclude that the $\rho - \theta$ data correspond to the generalized Archie model as expected, at least at a first approximation. The values of the fitting parameters are given in Table 1. Our future research will evaluate this relationship in order to verify it with more in-situ measurements from the twin niches, but also with additional laboratory experiment.

Once the empirical fit parameters of equation (4) are determined, the ERT data can be converted to estimates of water content. An example result is given in Fig. 7. Such water content images can directly be compared to the results of the numerical simulations.

5 Temporal Evolution of the Desaturation Process

In this section, the ERT measurements are interpreted statistically to provide time-dependent information about the resistivity change and the related water content evolution around the niches. Furthermore, the water content evolution

Fig. 8 Methodic approach: Concept of comparative multi-disciplinary investigation of the water content consisting of measurements and numerical modeling

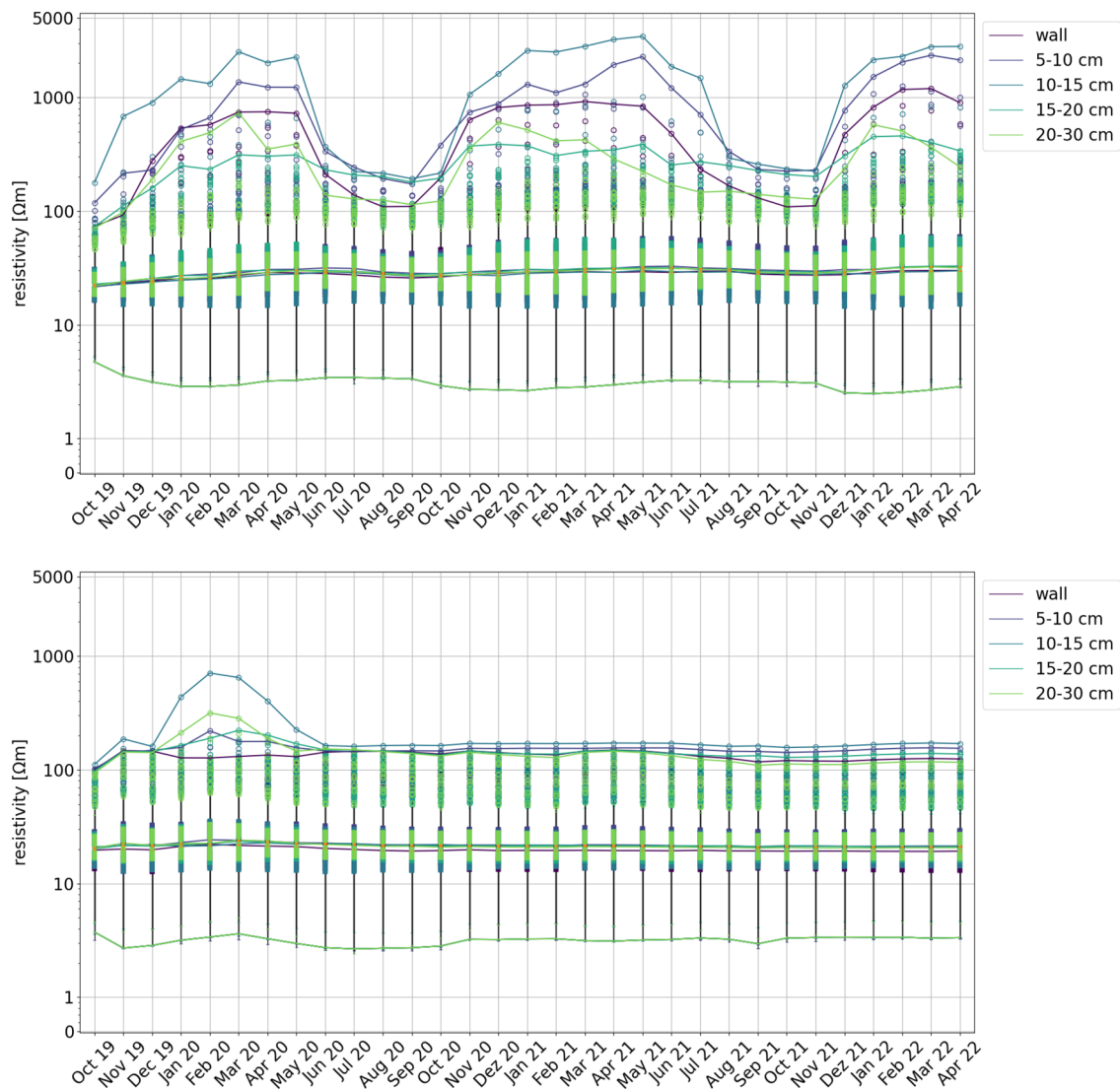
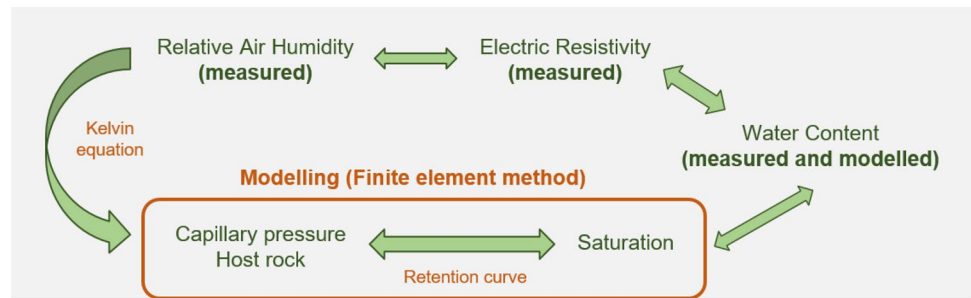


Fig. 9 Boxplot presentation of resistivity data set obtained experimentally for the open (top) and the closed niche (bottom)

is compared to numerical modelling results to discuss the modelling approaches. Aiming on a validation of the used numerical model we follow the methodic approach presented in Fig. 8, which indicates different methods to gain information about the saturation effects around the twin niches. While

the relative air humidity and the electrical resistivity can be derived by measurements, there are different ways to gain information about the water content. The NMR measurements aim at a direct measurement of the water content while the ERT measurements can be correlated to the water content

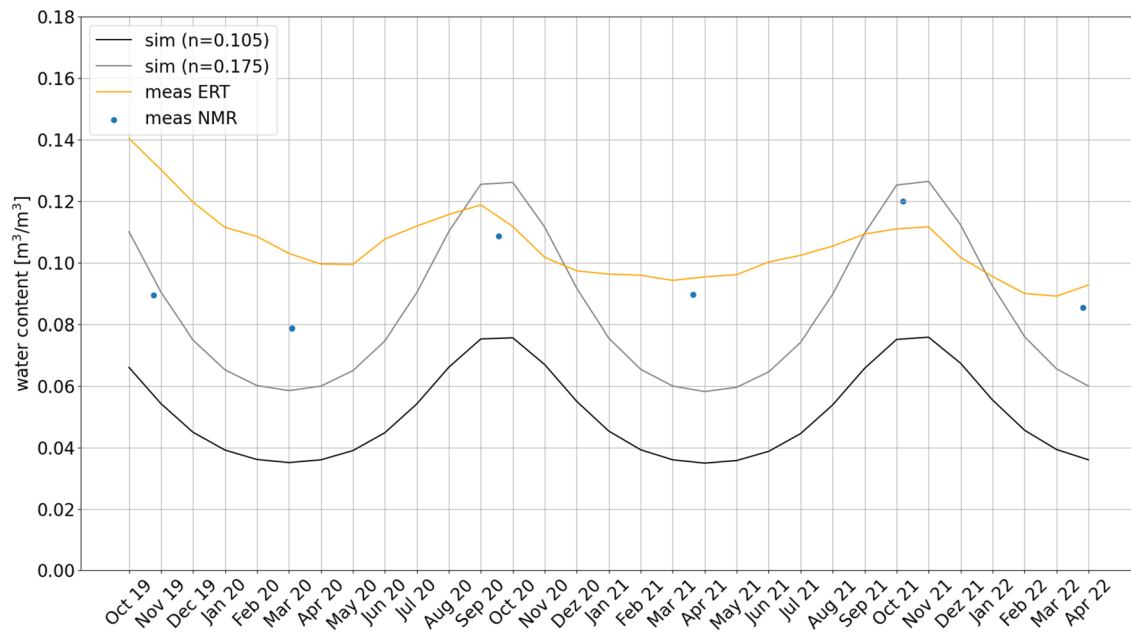


Fig. 10 Time-dependent evolution of water content on the wall inside the open niche as measured using NMR (blue points) as presented in Fig. 6 and estimated from measured mean resistivities (orange) as presented in Fig. 5 compared to the water content from the numerical simulation as presented in Fig. 4 with a porosity of $n = 0.105$ (darker

grey) and from a numerical simulation with maximum experimentally obtained porosity of $n = 0.175$ (lighter grey). The resistivity data was converted to values of water content by using the relation presented in Fig. 7

using the defined relation (described in Fig. 7). The numerical model calculates the water content based on balance equations and the defined constitutive relationships as they are defined in the model-set-up and applied in the finite element code OpenGeoSys indicated by the orange rectangle in Fig. 8. The measured relative air humidity is used as input parameter as introduced in Sect. 3. Uncertainties coming along with this approach have already been discussed in Sect. 3. Additional complexity arises due to heterogeneities, which are covered by the measurements and their statistical interpretation.

Conditioned by the relatively short measuring interval of 2 years of this long-term experiment, the quantitative results have to be interpreted critically aiming on further development of the methodical approach and future use of this approach for calibration and further development of the investigation methods. For this purpose, one measured ERT profile per month is evaluated statistically for each niche at different depths, i.e., distances from the niche wall as shown in Fig. 9. For that, an idealized ring profile with 2.4 m diameter is used and the measurements are re-computed with BERT. Since only the position of the sensors are shifted to the ideal niche contour, the interpolation process in BERT is not affected by the idealized contour.

Figure 9 shows the monthly temporal evolution of the resistivity from October 2019 to April 2022 for both niches, using boxplots, which are also called box-and-whisker plots as described in Wikipedia (2023). Each boxplot represents

the distribution of resistivity values on a virtual circular profile inside the niche wall. As is usual with boxplots, the boxes represent the interquartile range (*IQR*), which is bounded by the lower quartile $Q1$ (25th percentile) and the upper quartile $Q3$ (75th percentile). The plotted vertical lines represent the data below and above the mentioned quartiles. The lower range is computed with $Q1 - 1.5IQR$, while the upper with $Q3 + 1.5IQR$. These limits are marked by the lower and upper whiskers, respectively. The outliers are plotted as circles. It is important to notice that the minimum of the data set is larger than the calculated value of the lower whisker and for this reason no outlier is seen for low resistivities. On the other hand, especially for the open niche the maximum of the data set is larger than the calculated value of the upper whisker. For this reason there are outliers in the upper part of the graph. The related data sets are also summarized in Appendix.

Summarizing this, the ranges around the mean values, minimum and outliers under the minimum remain approximately constant and show only minor influence of seasonal effects. However, especially in the open twin the maximum and outliers above the maximum, i.e., at a higher resistivity ratio show a different behaviour. Here, significant seasonal variations over time occur with maximum amplitudes during winter time. This high range of measured resistivities indicates a significantly higher range of the water content along the niche profile, coming along with zones of a significant

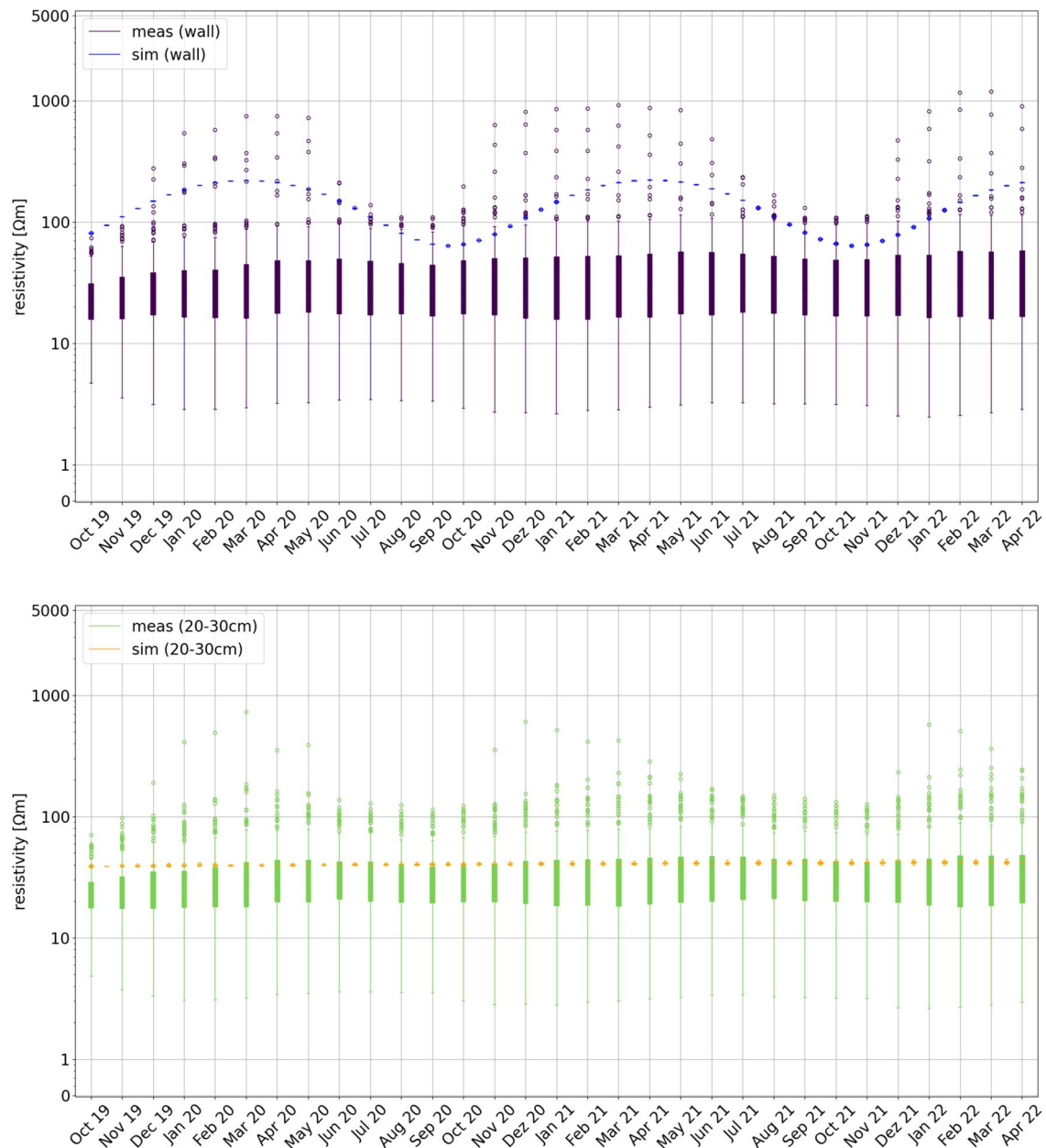


Fig. 11 Comparison of resistivity values obtained experimentally (in boxplot presentation mode (purple and green)) to resistivity values computed from simulated water content (given by dashed lines (blue and orange)). The simulated values are converted from saturation with

mathematical relation between resistivity and water content depicted in Fig. 7. The comparison is given near the surface (top) and in a depth of 20–30 cm (bottom)

increase of the resistivity or synonymously a significant decrease of the water content. This observation yields from the niche wall up to a depth of 20 to 30 cm which is the whole investigation area. The evaluation indicates that the outliers show a maximum amplitude at depths between 5 and 10 cm, nevertheless they occur in the whole measuring interval between 0 and 30 cm depths. The observed seasonal variations are interpreted as changes of the water content,

but they may also come along with a varying geometric structure or material parametrisation of possible cracks. The evolution of the measured elastic resistivity is also plotted in comparison to the modelled values directly at the niche wall and in a depth of 20 to 30 cm in Fig. 11.

Following the methodic approach presented in Fig. 8, we aim on a comparison of these measurements with the NMR measurements and the modelling results. For that, the

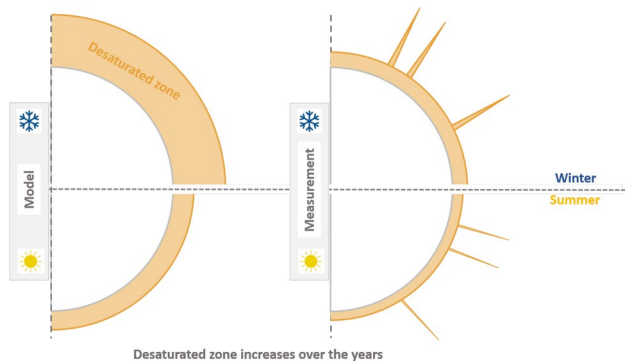


Fig. 12 Comparative schematic presentation of the desaturated zone and heterogeneities in the open twin resulting from the statistic evaluation of ERT data and the numerical model

derived $\rho - \theta$ -relationship (see Fig. 7) is applied to the complete ERT monitoring data set (Fig. 9), from which we get an impression of the temporal evolution of the mean water content round the niches. Figure 10 presents the corresponding data together with the NMR results and compares them with the results computed by numerical simulations assuming two different porosities for the OPA that vary within the experimentally obtained range Bossart et al. (2018). The curves indicate a principle agreement of the seasonal water content changes, although the numerical results indicate higher dynamics, i.e., higher amplitudes, of the seasonal variations. Furthermore, the geophysical measurements indicate a general decreasing trend over time, which superimposes the seasonal behaviour. The measured values are 1.5 to 3 times higher than the modelling results assuming a porosity of $n = 0.105$. Assuming a higher porosity of $n = 0.175$, the measured water contents are up to 2 times higher than the modelled values after the winter month and they correspond to the measurements at the end of the summer months. Both porosity values lie in the range of related laboratory measurements Bossart et al. (2018), i.e., $n = 0.105$ corresponds to the “best estimate” and $n = 0.175$ to the maximum experimentally obtained porosity. Ongoing use of the presented methodic approach will be based on a more comprehensive data set and will allow a statement as to whether the porosity of the sandy facies lies in the higher range, or is related to different lithological structures. The differences could also emerge from the modelling approach and its strong simplification, e.g., as already discussed when using the Kelvin equation. In general we can state, that the current numerical model obviously overestimates the desaturation effect.

For comparison and verification of the proposed methodic approach, the $\rho - \theta$ -relationship can also be used in the opposite direction, i.e., starting with the modelled water content values as input parameters to obtain electrical resistivity

estimates for the comparison with the measured ERT data. Consequently, the resulting water content is plugged in $1.108 \theta^{-1.58}$ (fitting from Fig. 7) to obtain the resistivity.

The simulated resistivity values are presented in Fig. 11 and compared with the ERT measurements. In analogy to Fig. 9, the results at the wall of the open niche and at the contour between 20 and 30 cm are plotted. We note that the modelled resistivities lay in the range of the ERT outliers, which, on the one hand, confirms the structural gap between numerical model and reality, but indicates, on the other hand, that the obtained values are in a physically meaningful range. In general, the derived values indicate much dryer conditions near the surface than the measurements. At higher depths of 20 to 30 cm, the model indicates low seasonal variations and similar values than the measurements. It can be stated, that the numerical model overestimates the measured mean resistivity and its seasonal variation near the niche wall, while it fits the measurements at higher depths quite good.

5.1 Discussion of the Methodology

Possible sources of error for the different results of numerical model and measurement can be found in each part of Fig. 8. First of all, measurements of high RH are crucial and imply lower accuracy. Furthermore, the position of the measurement can affect the results, as the humidity in the middle of the niches can differ from the situation at the surface. In the CD-A experiment, several measuring devices with best possible accuracy (please refer to Sect. 4) are used and positioned along the niches at different heights to keep the measurement error as low as possible. The behavior at the surface can also be influenced by material heterogeneities, the development of the EDZ and the fracture network. The related uncertainties can also be seen in the measurements and are part of the statistical evaluation presented here. Finally, osmotic effects may come into play. Osmotic pressure is primarily influenced by the concentration of dissolved solutes in water and the temperature. As Opalinus Clay is typically located in deep geological formations its pore water is usually relatively low in dissolved solutes Géologie (2003) and consequently osmotic effects are not a major factor in Opalinus Clay’s behavior. On the other hand, matric suction arises due to the attractive forces between the water molecules and the solid surfaces of the clay particles. As the moisture content decreases, the capillary forces increase, leading to higher matric suction. The low permeability and high retention properties of OPA prevent significant water flow through the clay over long timescales. In summary, in comparison to osmotic suction, matric suction seems to be the more critical factor in the water retention

behavior of Opalinus Clay Villar and Romero (2012), as it governs the capillary processes within the clay matrix and helps maintain a stable moisture content. Osmotic suction, while generally less significant, may come into play under certain conditions where there are notable differences in solute concentrations in the clay's surroundings. Nevertheless, the validity of the Kelvin equation must be critically questioned in this context. In general, the constitutive model as used in the numerical set-up might be too simple, e.g., the computation of the capillary pressure via Kelvin equation, fitting of the saturation via the van Genuchten retention curve or the Bishop's approximation for the effective stress. Additionally, the complexity of the claystone regarding the determination of its material parameters and water conductivity behaviour is a further aspect that leads to uncertainties. Due to the material structure of the claystone, effects like interfacial conductivity have to be kept in mind and make the interpretation of measured values very complex. Another important issue regarding this evaluation procedure is of course the reliability of the derived relationship between electrical resistivity and water content ($\rho - \theta$ -relationship), which will be in the focus of further investigations. The relatively short measuring period up to now comes along with high uncertainties in the empirical estimation of the fitting parameters of Eq. 4, which can hopefully be resolved by collecting more data over a longer time period during the ongoing experiment. Further evaluation of this long-term evolution might confirm this trend for all investigation methods.

As discussed in the previous section, the ERT outliers can reliably be associated with heterogeneities in the clay-rock (as depicted in Fig. 5), which are not incorporated in the numerical modeling approach. The homogeneous model approach, although delivering plausible qualitative results, is obviously limited regarding the quantitative differences to the geophysical observations. Further investigation is required to identify the influence of the measured heterogeneities and micro-cracks on the hydraulic as well as mechanical properties in the numerical model. Depending on the scale and also the sealing behaviour, the small scale effects may have an impact on the larger scale due to local gas pressure built up or potential ongoing damage, which might be of significant importance concerning the integrity of a potential waste repository. Consequently, heterogeneous effects such as cracks indicated by systematically appearing outliers in the measurements and the local influence of these cracks on the drying behaviour, need to be taken into account Cajuhi et al. (2018, 2021a, 2023b). A schematic presentation of

the discussed results is summarized in Fig. 12 indicating the overestimation of the desaturated zone in the modelling approach and the appearance of cracks observed by the measurements. The difference between model and measurement becomes even more obvious in winter than in summer time.

Finally, it has to be pointed out that the presented methodical approach to combine measurements and modelling approach has to be assessed, tested and further developed on the basis of the long-term evaluation of the CD-A experiment, which results in a comprehensive data set within the next years. Nevertheless, the presented approach provides an opportunity for more process understanding and enables a calibration of various measuring as well as modelling strategies, enabling continued development and statements concerning possible uncertainties.

6 Conclusion and Outlook

With the presented multi-disciplinary methods we investigate the characteristics of the claystone and the evolution of the water content around the twin niches, which have a significant impact on coupled HM behavior in claystone. The continual, ongoing ERT measurements provide a comprehensive data set for the evolution of the electrical resistivity and reflect clearly the climatic conditions in the niches. This data set is interpreted by means of a statistical evaluation. The data points can be assigned to two groups with different characteristics: While about 90% of the 72 sensor locations indicate a change of resistivities plotting in a narrow range, there exist a second group with outliers, showing significantly seasonal behaviour, which correlates to the relative air humidity in the niche. Interpreting this seasonally influenced zones, we assume the existence of small-scale singularities in contrast to the dominating, more homogeneous zone. These seasonally influenced zones often correlate with open fractures, whereas carbonate and sand lenses as well as shear zones or scaly clay tend to have minor impact. Furthermore, already the initial measurement allows for an assessment of seasonally affected zones, as high initial resistivities often correspond with zones with a strong increase of resistivity over time. The seasonalities in the ERT measurements indicate zones with particularly pronounced hydraulic behaviour coming along with higher permeabilities, thus, the containment properties of the clay rock are here affected.

In combination with the seasonal NMR in-situ measurements on the tunnel wall, a preliminary relationship between electrical resistivity and water content can be established that enables information about the evolution of water content around the excavations. This information yields in a time dependent moisture map and might be a good basis for an estimated permeability evolution and comparison with numerical modelling. However, this relationship is still vague due to the few data sets at the beginning of the CD-A experiment, but future work in the long-term should underline and consolidate the presented trend. Due to the mentioned high level of material heterogeneity in the near field, the set-up and calibration of a model approach is a challenge with different possibilities to handle. Small scale modeling like e.g., phase field approaches may represent the fracture evolution and enable basic information for statements about permeability evolution, sealing, and related effects like gas transport. This kind of modelling may be calibrated and also validated by means of the presented methodic approach using a combined interpretation of measurements and modelling.

Nevertheless, in case of a potential repository, the input parameters for such a detailed modelling approach are rare. However, a sound understanding of the mentioned small scale effects serves as an important basis for the establishment of a pragmatic, potentially homogenized approach to handle near field, long-term, heterogeneous, coupled HM effects in the safety assessment of a potential repository for HLW. The long-term CD-A experiment is a favorable environment to further evaluate the presented investigation procedure aiming on a quantitative evaluation and verification of the used measuring and modelling approaches.

Data sets for the Statistical Evaluation of Resistivity as Discussed in Sect. 5

For completeness, the data sets of the boxplots as presented in Sect. 5 are summarized here in tabular form for different depths in Tables 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11.

Table 2 Resistivity data set of the wall of the open niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	4.73	24.63	73.45	15.76	31.02	15.26	- 7.13	53.9
Nov 19	3.57	27.74	92.33	15.95	35.36	19.41	- 13.17	64.48
Dec 19	3.13	31.69	277.05	17.1	38.31	21.21	- 14.72	70.13
Jan 20	2.87	35.47	538.29	16.38	39.99	23.61	- 19.04	75.41
Feb 20	2.87	37.01	574.79	16.3	40.31	24.0	- 19.7	76.31
Mar 20	2.96	40.19	742.34	16.07	44.85	28.77	- 27.09	88.01
Apr 20	3.21	42.43	746.32	17.68	47.94	30.26	- 27.7	93.32
May 20	3.25	42.51	725.4	18.01	48.02	30.01	- 27.01	93.04
Jun 20	3.43	37.47	211.32	17.5	49.67	32.17	- 30.75	97.92
Jul 20	3.43	35.26	138.01	17.17	47.58	30.41	- 28.46	93.2
Aug 20	3.39	33.47	109.32	17.6	45.86	28.26	- 24.79	88.25
Sep 20	3.34	32.08	109.8	16.8	44.19	27.39	- 24.3	85.28
Oct 20	2.92	35.32	195.97	17.55	47.91	30.36	- 27.99	93.45
Nov 20	2.7	40.99	633.21	17.13	49.94	32.81	- 32.08	99.15
Dez 20	2.68	43.95	813.91	16.09	50.63	34.54	- 35.73	102.45
Jan 21	2.64	44.73	853.64	15.78	51.64	35.86	- 38.01	105.43
Feb 21	2.8	44.99	861.51	15.86	52.26	36.4	- 38.74	106.86
Mar 21	2.84	46.26	920.04	16.43	52.88	36.46	- 38.26	107.57
Apr 21	2.97	45.38	871.33	16.46	54.7	38.24	- 40.9	112.07
May 21	3.12	44.85	836.74	17.44	56.79	39.35	- 41.58	115.82
Jun 21	3.25	41.98	479.83	17.08	56.21	39.14	- 41.63	114.92
Jul 21	3.26	40.57	235.07	18.0	54.56	36.55	- 36.82	109.39
Aug 21	3.16	38.77	166.68	17.72	52.17	34.45	- 33.95	103.84
Sep 21	3.17	36.58	130.7	17.14	49.49	32.35	- 31.38	98.02
Oct 21	3.13	35.73	108.87	16.86	48.79	31.93	- 31.04	96.69
Nov 21	3.07	35.38	111.38	16.77	49.12	32.34	- 31.74	97.63
Dez 21	2.52	41.0	472.32	17.07	53.28	36.21	- 37.25	107.6
Jan 22	2.48	45.31	819.92	16.28	53.16	36.88	- 39.03	108.47
Feb 22	2.55	49.75	1172.09	16.59	57.33	40.74	- 44.52	118.43
Mar 22	2.67	50.54	1194.71	16.02	57.04	41.02	- 45.51	118.56
Apr 22	2.86	47.45	898.91	16.65	58.25	41.6	- 45.75	120.65

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 3 Resistivity data set concerning the 5–10 cm depth of the open niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	4.74	26.37	117.87	17.04	32.1	15.06	– 5.56	54.7
Nov 19	3.56	30.76	214.95	16.6	35.24	18.64	– 11.37	63.21
Dec 19	3.11	34.26	229.99	16.11	40.61	24.5	– 20.64	77.36
Jan 20	2.87	39.52	521.86	15.92	44.46	28.54	– 26.88	87.26
Feb 20	2.87	41.63	662.63	15.9	44.53	28.63	– 27.05	87.48
Mar 20	2.96	48.93	1366.55	16.23	48.57	32.34	– 32.28	97.09
Apr 20	3.19	49.73	1229.18	17.55	51.7	34.15	– 33.68	102.92
May 20	3.23	48.53	1225.12	18.35	53.04	34.68	– 33.67	105.07
Jun 20	3.37	42.52	335.0	18.68	51.03	32.35	– 29.84	99.55
Jul 20	3.35	39.56	242.94	18.39	49.0	30.61	– 27.53	94.91
Aug 20	3.33	36.98	193.1	18.23	45.71	27.48	– 22.99	86.92
Sep 20	3.29	35.25	173.37	18.2	43.9	25.7	– 20.35	82.45
Oct 20	2.8	38.83	379.55	18.16	45.27	27.11	– 22.5	85.93
Nov 20	2.66	43.88	739.89	17.89	50.22	32.33	– 30.61	98.73
Dez 20	2.66	48.85	878.13	17.52	55.57	38.05	– 39.56	112.66
Jan 21	2.62	53.91	1305.9	16.5	58.06	41.57	– 45.85	120.41
Feb 21	2.78	54.31	1097.85	16.86	57.64	40.78	– 44.31	118.81
Mar 21	2.82	57.13	1310.85	17.0	58.67	41.67	– 45.51	121.17
Apr 21	2.95	60.08	1936.17	17.37	57.98	40.61	– 43.55	118.9
May 21	3.09	59.96	2284.43	18.55	60.0	41.45	– 43.62	122.17
Jun 21	3.17	52.25	1216.02	18.85	59.91	41.06	– 42.74	121.5
Jul 21	3.08	48.67	708.26	19.32	57.97	38.65	– 38.65	115.94
Aug 21	2.84	44.62	335.32	19.21	54.85	35.64	– 34.24	108.3
Sep 21	2.88	41.11	232.42	18.87	51.44	32.57	– 29.98	100.3
Oct 21	2.89	40.01	224.71	18.76	49.68	30.92	– 27.62	96.06
Nov 21	2.86	39.59	229.45	18.62	49.28	30.66	– 27.37	95.27
Dez 21	2.48	47.01	763.8	17.33	55.82	38.49	– 40.41	113.56
Jan 22	2.47	54.62	1514.54	17.0	58.28	41.28	– 44.93	120.21
Feb 22	2.52	61.86	2042.9	16.93	60.6	43.67	– 48.57	126.1
Mar 22	2.65	65.56	2346.44	17.02	62.36	45.35	– 51.0	130.38
Apr 22	2.82	62.51	2132.05	17.98	62.24	44.26	– 48.42	128.64

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 4 Resistivity data set concerning the 10–15 cm depth of the open niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	5.3	25.58	177.65	15.74	32.08	16.34	– 8.78	56.59
Nov 19	4.23	31.34	680.92	14.45	35.44	20.99	– 17.04	66.92
Dec 19	3.86	35.16	902.27	14.65	41.22	26.57	– 25.2	81.08
Jan 20	3.42	40.77	1450.12	14.28	44.06	29.78	– 30.4	88.74
Feb 20	3.43	41.71	1320.34	14.17	43.38	29.21	– 29.64	87.19
Mar 20	3.51	50.96	2515.08	14.03	46.83	32.81	– 35.19	96.05
Apr 20	3.86	48.88	2014.96	14.14	50.6	36.46	– 40.55	105.29
May 20	3.96	49.77	2262.92	14.69	52.04	37.36	– 41.35	108.08
Jun 20	3.91	38.28	370.39	15.45	49.19	33.73	– 35.15	99.79
Jul 20	3.02	35.37	223.02	15.46	44.66	29.21	– 28.36	88.48
Aug 20	3.52	33.25	215.46	15.39	41.33	25.94	– 23.53	80.25
Sep 20	3.38	31.76	192.67	15.12	38.97	23.85	– 20.66	74.75
Oct 20	3.18	33.96	216.05	14.76	42.26	27.49	– 26.48	83.5
Nov 20	3.15	41.43	1059.75	13.98	46.54	32.56	– 34.87	95.38
Dez 20	3.3	46.97	1610.92	14.39	50.24	35.85	– 39.39	104.02
Jan 21	3.21	54.09	2583.82	13.95	51.04	37.09	– 41.68	106.68
Feb 21	3.37	54.39	2504.92	14.06	52.36	38.31	– 43.41	109.83
Mar 21	3.44	57.22	2811.69	14.09	52.89	38.79	– 44.1	111.08
Apr 21	3.61	59.37	3227.05	14.32	51.47	37.14	– 41.39	107.18
May 21	3.65	59.28	3441.46	14.59	53.85	39.25	– 44.28	112.73
Jun 21	3.6	49.93	1870.54	14.92	54.86	39.93	– 44.97	114.75
Jul 21	3.45	47.13	1478.84	14.96	52.44	37.48	– 41.25	108.65
Aug 21	3.14	39.1	292.56	15.66	48.89	33.24	– 34.2	98.75
Sep 21	3.22	36.48	258.98	15.49	44.51	29.02	– 28.03	88.03
Oct 21	3.25	35.54	235.29	15.35	43.24	27.89	– 26.48	85.08
Nov 21	3.23	35.08	224.23	15.27	42.95	27.68	– 26.25	84.47
Dez 21	2.91	44.08	1265.48	13.93	49.42	35.49	– 39.3	102.65
Jan 22	3.05	51.5	2144.16	13.58	53.25	39.67	– 45.93	112.76
Feb 22	3.16	55.5	2292.78	14.37	54.08	39.71	– 45.2	113.65
Mar 22	3.28	59.49	2789.76	13.84	54.79	40.95	– 47.59	116.22
Apr 22	3.39	58.51	2802.22	14.54	55.6	41.06	– 47.05	117.19

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 5 Resistivity data set concerning the 15–20 cm depth of the open niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	5.13	25.98	72.81	18.52	30.97	12.45	− 0.15	49.64
Nov 19	4.04	29.32	111.33	18.28	35.94	17.66	− 8.2	62.43
Dec 19	3.65	32.33	159.18	18.88	39.51	20.63	− 12.07	70.46
Jan 20	3.26	35.94	250.68	18.76	45.17	26.42	− 20.87	84.8
Feb 20	3.26	37.29	233.05	18.86	46.91	28.05	− 23.22	88.98
Mar 20	3.34	42.2	312.16	19.13	50.93	31.8	− 28.57	98.63
Apr 20	3.67	42.37	303.87	19.87	52.35	32.48	− 28.84	101.07
May 20	3.76	42.82	311.52	20.8	53.49	32.69	− 28.24	102.53
Jun 20	4.08	39.06	230.87	20.64	49.21	28.57	− 22.21	92.07
Jul 20	4.13	36.82	207.38	21.01	45.67	24.66	− 15.99	82.67
Aug 20	4.06	35.02	200.27	21.22	42.02	20.8	− 9.98	73.21
Sep 20	3.98	33.63	179.76	20.97	39.82	18.85	− 7.31	68.1
Oct 20	3.56	35.84	194.72	20.65	42.38	21.72	− 11.94	74.96
Nov 20	3.23	39.97	372.61	20.43	47.93	27.49	− 20.81	89.17
Dez 20	3.13	43.43	386.47	19.83	53.58	33.76	− 30.81	104.22
Jan 21	3.06	46.21	375.3	19.23	56.46	37.23	− 36.62	112.31
Feb 21	3.22	46.4	307.55	19.15	57.14	37.99	− 37.84	114.13
Mar 21	3.28	47.75	335.53	19.02	58.92	39.9	− 40.83	118.76
Apr 21	3.44	48.26	344.87	19.92	57.75	37.84	− 36.84	114.51
May 21	3.65	48.42	387.09	20.37	57.69	37.32	− 35.6	113.66
Jun 21	3.87	44.91	254.01	20.58	54.18	33.6	− 29.81	104.57
Jul 21	3.94	43.06	271.56	20.02	51.88	31.86	− 27.76	99.67
Aug 21	3.96	40.56	251.03	21.01	48.87	27.86	− 20.79	90.66
Sep 21	3.92	38.24	226.92	21.03	45.57	24.54	− 15.78	82.37
Oct 21	3.85	37.2	209.65	21.15	44.01	22.86	− 13.13	78.29
Nov 21	3.77	36.71	201.2	21.07	43.88	22.81	− 13.15	78.1
Dez 21	3.04	42.13	304.75	20.16	49.44	29.28	− 23.77	93.37
Jan 22	2.9	47.39	451.14	19.53	57.63	38.1	− 37.62	114.77
Feb 22	3.0	49.94	457.5	19.35	61.93	42.57	− 44.51	125.79
Mar 22	3.14	51.53	400.88	19.67	60.81	41.14	− 42.05	122.53
Apr 22	3.35	50.26	338.97	20.06	59.58	39.52	− 39.21	118.85

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 6 Resistivity data set concerning the 20–30 cm depth of the open niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	4.86	24.64	70.89	17.69	29.04	11.35	0.66	46.07
Nov 19	3.74	27.47	97.93	17.58	32.02	14.44	– 4.07	53.67
Dec 19	3.35	30.48	189.48	17.5	35.24	17.74	– 9.1	61.84
Jan 20	3.06	33.25	409.54	17.9	35.59	17.69	– 8.64	62.12
Feb 20	3.11	34.84	488.89	18.02	38.24	20.22	– 12.31	68.58
Mar 20	3.22	39.33	730.64	18.12	42.24	24.12	– 18.06	78.41
Apr 20	3.42	37.99	350.47	19.88	43.64	23.76	– 15.75	79.27
May 20	3.47	38.48	388.96	19.9	43.94	24.04	– 16.15	80.0
Jun 20	3.59	36.03	138.0	20.97	42.55	21.58	– 11.41	74.92
Jul 20	3.59	34.86	127.95	20.08	42.35	22.28	– 13.34	75.77
Aug 20	3.56	33.34	124.5	19.69	39.71	20.02	– 10.34	69.74
Sep 20	3.53	32.26	114.0	19.36	38.75	19.39	– 9.72	67.83
Oct 20	3.05	34.12	122.39	19.9	39.18	19.28	– 9.02	68.09
Nov 20	2.84	36.76	355.28	19.9	40.68	20.78	– 11.28	71.85
Dez 20	2.86	39.81	602.39	19.19	42.74	23.56	– 16.15	78.08
Jan 21	2.81	41.15	518.35	18.42	44.01	25.59	– 19.97	82.4
Feb 21	2.96	41.09	414.91	18.65	44.07	25.43	– 19.49	82.21
Mar 21	3.03	42.06	423.96	18.18	44.75	26.58	– 21.69	84.62
Apr 21	3.12	41.68	287.17	19.13	45.48	26.35	– 20.39	85.01
May 21	3.25	41.03	224.32	19.66	46.74	27.08	– 20.96	87.36
Jun 21	3.39	40.2	170.05	19.99	46.98	27.0	– 20.51	87.48
Jul 21	3.42	39.36	146.6	20.77	46.52	25.75	– 17.86	85.16
Aug 21	3.26	38.16	149.89	21.09	44.78	23.69	– 14.44	80.32
Sep 21	3.25	36.55	140.52	20.33	44.7	24.37	– 16.22	81.25
Oct 21	3.22	35.63	132.14	19.98	42.71	22.73	– 14.11	76.8
Nov 21	3.16	35.17	127.07	19.93	41.17	21.23	– 11.92	73.01
Dez 21	2.67	38.56	233.19	19.7	43.27	23.57	– 15.66	78.63
Jan 22	2.63	42.37	574.16	18.74	44.56	25.82	– 19.99	83.3
Feb 22	2.7	44.25	506.93	18.17	47.56	29.39	– 25.92	91.65
Mar 22	2.81	44.57	364.6	18.46	47.69	29.23	– 25.39	91.54
Apr 22	2.95	43.54	245.05	19.46	48.09	28.63	– 23.48	91.03

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5IQR$ and $Q3 + 1.5IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 7 Resistivity data set of the wall of the closed niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	3.74	24.6	99.75	12.91	29.42	16.52	- 11.87	54.2
Nov 19	2.71	27.95	147.41	12.25	34.08	21.82	- 20.48	66.81
Dec 19	2.86	27.1	145.95	12.17	31.09	18.92	- 16.2	59.46
Jan 20	3.17	28.51	127.42	12.66	33.95	21.3	- 19.29	65.89
Feb 20	3.38	29.93	127.23	13.61	35.7	22.09	- 19.53	68.84
Mar 20	3.61	29.4	130.95	13.21	35.84	22.64	- 20.75	69.79
Apr 20	3.27	28.27	134.78	13.45	33.2	19.75	- 16.19	62.83
May 20	2.97	27.7	130.57	13.11	31.61	18.5	- 14.64	59.37
Jun 20	2.73	27.29	142.82	12.91	31.57	18.65	- 15.07	59.55
Jul 20	2.66	26.99	145.21	12.78	30.33	17.55	- 13.55	56.66
Aug 20	2.7	26.82	146.16	12.87	30.13	17.26	- 13.02	56.03
Sep 20	2.72	26.56	142.82	12.84	30.14	17.29	- 13.09	56.07
Oct 20	2.82	26.38	137.55	13.04	30.21	17.17	- 12.71	55.96
Nov 20	3.23	26.56	146.51	12.73	30.4	17.68	- 13.79	56.92
Dez 20	3.22	26.4	141.62	12.76	30.23	17.46	- 13.43	56.42
Jan 21	3.26	26.36	137.52	12.73	30.54	17.8	- 13.97	57.24
Feb 21	3.26	26.27	135.2	12.71	30.24	17.53	- 13.58	56.53
Mar 21	3.15	26.76	146.26	13.06	30.33	17.28	- 12.86	56.25
Apr 21	3.12	26.9	149.36	13.15	30.17	17.03	- 12.4	55.72
May 21	3.19	26.58	146.47	12.99	30.17	17.18	- 12.79	55.94
Jun 21	3.22	26.31	139.48	12.74	30.12	17.38	- 13.32	56.19
Jul 21	3.32	26.01	131.18	12.69	29.99	17.29	- 13.25	55.93
Aug 21	3.24	25.66	126.11	12.85	29.3	16.45	- 11.82	53.97
Sep 21	2.96	25.43	117.34	13.09	29.15	16.06	- 11.0	53.24
Oct 21	3.31	25.42	120.58	12.92	29.09	16.17	- 11.33	53.35
Nov 21	3.35	25.44	119.44	12.87	29.37	16.5	- 11.88	54.12
Dez 21	3.38	25.51	118.91	12.82	29.65	16.83	- 12.43	54.89
Jan 22	3.36	25.57	122.26	12.64	29.81	17.17	- 13.11	55.57
Feb 22	3.37	25.63	124.71	12.57	29.61	17.04	- 12.98	55.16
Mar 22	3.31	25.59	125.65	12.47	29.5	17.03	- 13.08	55.05
Apr 22	3.34	25.65	124.58	12.49	29.74	17.25	- 13.39	55.61

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 8 Resistivity data set concerning the 5–10 cm depth of the closed niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	3.2	25.19	101.99	15.21	29.84	14.63	– 6.73	51.79
Nov 19	2.71	29.28	142.91	15.29	35.08	19.79	– 14.39	64.76
Dec 19	2.87	28.51	148.19	15.1	34.1	19.0	– 13.4	62.61
Jan 20	2.95	30.52	157.35	15.98	34.7	18.72	– 12.09	62.77
Feb 20	3.14	32.13	219.63	16.06	35.64	19.57	– 13.3	65.0
Mar 20	3.23	31.61	177.21	16.64	35.91	19.28	– 12.28	64.83
Apr 20	2.93	30.72	177.64	15.91	34.98	19.08	– 12.71	63.6
May 20	2.75	29.79	156.35	15.52	34.88	19.36	– 13.52	63.91
Jun 20	2.59	29.16	146.55	15.43	33.6	18.18	– 11.84	60.87
Jul 20	2.55	28.52	143.88	14.9	33.0	18.1	– 12.24	60.14
Aug 20	2.55	28.01	146.21	14.8	32.06	17.27	– 11.1	57.96
Sep 20	2.56	27.56	147.0	14.58	31.5	16.91	– 10.79	56.87
Oct 20	2.63	27.21	146.25	14.38	31.51	17.14	– 11.33	57.22
Nov 20	3.19	27.55	154.5	14.31	31.35	17.04	– 11.24	56.9
Dez 20	3.16	27.27	153.75	14.15	31.2	17.05	– 11.41	56.77
Jan 21	3.17	27.13	154.69	14.14	31.54	17.41	– 11.97	57.65
Feb 21	3.18	26.97	154.43	14.11	31.52	17.41	– 12.01	57.64
Mar 21	3.09	27.77	154.3	14.32	32.4	18.08	– 12.81	59.53
Apr 21	3.09	28.04	155.55	14.16	32.58	18.43	– 13.48	60.22
May 21	3.16	27.52	155.6	14.09	31.73	17.64	– 12.37	58.19
Jun 21	3.16	27.09	155.54	14.06	31.37	17.31	– 11.91	57.34
Jul 21	3.18	26.63	150.46	14.1	30.99	16.89	– 11.24	56.33
Aug 21	3.07	26.24	145.49	14.02	30.22	16.2	– 10.27	54.51
Sep 21	2.68	25.73	144.75	14.26	29.99	15.73	– 9.35	53.59
Oct 21	3.08	25.91	142.05	14.38	29.71	15.33	– 8.6	52.7
Nov 21	3.11	25.9	143.88	14.22	29.65	15.43	– 8.92	52.79
Dez 21	3.17	25.96	147.08	13.93	29.76	15.83	– 9.82	53.51
Jan 22	3.22	26.01	151.52	13.98	30.2	16.22	– 10.35	54.53
Feb 22	3.27	26.07	154.81	14.05	30.23	16.19	– 10.23	54.51
Mar 22	3.23	26.04	155.96	14.03	30.26	16.23	– 10.32	54.61
Apr 22	3.24	26.11	154.6	14.0	30.61	16.61	– 10.91	55.52

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 9 Resistivity data set concerning the 10–15 cm depth of the closed niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	3.59	24.53	111.42	13.53	28.87	15.34	– 9.48	51.88
Nov 19	2.68	28.25	187.01	12.28	33.26	20.98	– 19.19	64.74
Dec 19	2.81	27.57	160.55	12.62	31.21	18.59	– 15.26	59.09
Jan 20	3.12	30.3	437.03	12.57	33.46	20.89	– 18.76	64.79
Feb 20	3.34	32.68	708.64	13.06	36.87	23.82	– 22.67	72.6
Mar 20	3.48	32.86	647.63	12.7	36.62	23.92	– 23.18	72.5
Apr 20	3.22	30.75	403.49	12.69	34.67	21.98	– 20.29	67.64
May 20	2.92	29.16	226.02	12.5	33.21	20.71	– 18.56	64.28
Jun 20	2.68	28.34	163.14	12.19	31.43	19.24	– 16.68	60.29
Jul 20	2.61	27.76	160.81	12.23	31.19	18.96	– 16.21	59.63
Aug 20	2.64	27.34	163.28	12.55	30.73	18.18	– 14.72	58.0
Sep 20	2.66	26.98	164.23	12.47	30.04	17.58	– 13.9	56.41
Oct 20	2.76	26.72	163.36	12.57	29.37	16.81	– 12.64	54.58
Nov 20	3.17	27.01	170.51	13.09	29.48	16.39	– 11.5	54.07
Dez 20	3.16	26.79	169.76	13.44	29.53	16.1	– 10.71	53.68
Jan 21	3.2	26.69	170.4	13.5	29.5	16.01	– 10.52	53.52
Feb 21	3.21	26.56	170.11	13.54	29.34	15.8	– 10.15	53.04
Mar 21	3.09	27.26	170.54	13.11	29.92	16.81	– 12.1	55.14
Apr 21	3.05	27.41	171.94	13.14	30.02	16.88	– 12.19	55.35
May 21	3.13	26.98	171.7	13.37	29.32	15.95	– 10.56	53.25
Jun 21	3.16	26.62	171.42	13.48	29.4	15.93	– 10.41	53.29
Jul 21	3.26	26.22	165.87	13.4	29.16	15.76	– 10.23	52.8
Aug 21	3.18	25.85	160.93	13.17	28.92	15.76	– 10.47	52.56
Sep 21	2.9	25.48	162.26	13.33	27.88	14.55	– 8.49	49.71
Oct 21	3.25	25.54	157.03	13.25	28.46	15.22	– 9.58	51.29
Nov 21	3.29	25.52	158.76	13.31	28.32	15.01	– 9.2	50.84
Dez 21	3.32	25.58	161.96	13.53	28.36	14.83	– 8.72	50.61
Jan 22	3.31	25.67	166.92	13.7	28.45	14.75	– 8.43	50.58
Feb 22	3.32	25.72	170.31	13.75	28.36	14.61	– 8.16	50.27
Mar 22	3.26	25.71	172.11	13.7	28.02	14.33	– 7.8	49.52
Apr 22	3.29	25.76	170.23	13.7	28.43	14.73	– 8.39	50.52

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 10 Resistivity data set concerning the 15–20 cm depth of the closed niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	4.47	25.52	97.26	15.78	28.25	12.47	– 2.93	46.96
Nov 19	3.98	29.53	145.47	14.89	33.0	18.11	– 12.28	60.17
Dec 19	4.0	28.78	144.2	16.16	30.56	14.4	– 5.44	52.16
Jan 20	4.24	30.02	164.14	15.08	32.95	17.87	– 11.72	59.75
Feb 20	4.54	31.56	189.14	14.68	34.63	19.95	– 15.25	64.55
Mar 20	4.55	31.54	222.23	15.25	33.79	18.54	– 12.56	61.6
Apr 20	4.42	30.75	201.83	15.59	31.82	16.23	– 8.76	56.16
May 20	4.03	30.15	169.24	15.75	31.25	15.5	– 7.49	54.49
Jun 20	3.79	29.72	149.54	16.3	29.61	13.31	– 3.67	49.58
Jul 20	3.72	29.13	149.34	15.84	28.97	13.13	– 3.86	48.67
Aug 20	3.66	28.64	145.67	15.41	28.57	13.15	– 4.32	48.29
Sep 20	3.68	28.18	141.15	15.18	28.23	13.05	– 4.4	47.81
Oct 20	3.81	27.75	134.6	15.2	28.27	13.07	– 4.39	47.87
Nov 20	4.38	27.95	146.41	15.1	28.5	13.4	– 5.0	48.6
Dez 20	4.4	27.64	140.33	15.1	28.32	13.22	– 4.73	48.16
Jan 21	4.46	27.43	137.66	15.05	28.24	13.18	– 4.72	48.01
Feb 21	4.49	27.25	137.46	15.07	28.19	13.12	– 4.62	47.88
Mar 21	4.27	28.16	145.5	15.21	28.61	13.41	– 4.9	48.72
Apr 21	4.21	28.49	148.77	15.16	28.56	13.4	– 4.93	48.65
May 21	4.34	27.95	145.3	14.99	28.32	13.34	– 5.02	48.33
Jun 21	4.38	27.45	138.36	14.86	28.13	13.27	– 5.05	48.04
Jul 21	4.55	26.91	134.53	15.04	27.99	12.94	– 4.37	47.4
Aug 21	4.42	26.52	131.31	15.14	27.61	12.47	– 3.56	46.3
Sep 21	4.07	26.07	132.99	15.29	27.16	11.87	– 2.52	44.97
Oct 21	4.5	26.13	128.51	15.3	27.34	12.04	– 2.76	45.41
Nov 21	4.57	26.07	129.64	15.29	27.24	11.95	– 2.64	45.17
Dez 21	4.64	26.09	131.81	15.34	27.23	11.89	– 2.49	45.07
Jan 22	4.7	26.13	135.38	15.18	27.31	12.13	– 3.02	45.5
Feb 22	4.75	26.17	137.77	15.28	27.23	11.95	– 2.65	45.16
Mar 22	4.68	26.14	139.35	15.26	27.19	11.92	– 2.62	45.08
Apr 22	4.7	26.2	137.85	15.2	26.95	11.75	– 2.42	44.58

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

Table 11 Resistivity data set concerning the 20–30 cm depth of the closed niche as defined in Sect. 5

Month	Min	Mean	Max	Q1	Q3	IQR	Q1-1.5IQR	Q3+ 1.5IQR
Oct 19	4.61	24.88	93.69	16.14	27.15	11.01	− 0.38	43.67
Nov 19	3.75	29.03	143.29	15.29	31.43	16.14	− 8.93	55.65
Dec 19	3.93	28.0	141.85	15.78	30.3	14.52	− 5.99	52.08
Jan 20	4.23	30.18	211.47	16.15	31.84	15.7	− 7.4	55.38
Feb 20	4.69	32.03	316.34	15.99	34.49	18.5	− 11.77	62.24
Mar 20	5.02	32.08	283.64	15.84	33.76	17.92	− 11.04	60.64
Apr 20	4.02	30.49	188.73	17.04	30.5	13.46	− 3.14	50.68
May 20	3.46	29.4	146.02	16.84	29.76	12.92	− 2.54	49.14
Jun 20	2.63	28.89	152.24	16.21	29.54	13.33	− 3.77	49.53
Jul 20	2.42	28.24	150.04	16.3	29.36	13.05	− 3.28	48.94
Aug 20	2.83	27.7	144.33	16.38	29.33	12.95	− 3.04	48.75
Sep 20	2.85	27.24	139.07	16.28	29.24	12.96	− 3.17	48.68
Oct 20	3.35	26.84	132.22	16.25	28.85	12.6	− 2.65	47.74
Nov 20	3.91	27.12	142.46	15.83	29.72	13.89	− 5.01	50.55
Dez 20	4.35	26.82	136.34	15.9	29.12	13.22	− 3.94	48.96
Jan 21	4.41	26.61	131.32	16.02	28.73	12.71	− 3.04	47.79
Feb 21	4.42	26.42	128.0	16.14	28.44	12.29	− 2.3	46.88
Mar 21	4.28	27.39	142.17	15.93	29.44	13.51	− 4.33	49.71
Apr 21	4.23	27.68	145.72	16.12	29.82	13.7	− 4.42	50.36
May 21	4.3	27.11	141.46	16.1	29.1	13.0	− 3.4	48.6
Jun 21	4.35	26.59	132.92	15.94	28.59	12.65	− 3.04	47.57
Jul 21	3.91	26.12	123.44	16.08	28.18	12.1	− 2.08	46.34
Aug 21	3.55	25.76	118.89	15.94	27.94	12.0	− 2.06	45.95
Sep 21	3.3	25.24	109.33	16.11	27.51	11.4	− 0.99	44.6
Oct 21	3.43	25.42	112.85	15.92	27.96	12.05	− 2.15	46.03
Nov 21	3.56	25.36	111.25	15.97	27.97	12.0	− 2.02	45.96
Dez 21	3.84	25.37	111.26	15.95	28.03	12.08	− 2.17	46.15
Jan 22	4.55	25.36	114.46	15.97	27.96	12.0	− 2.03	45.96
Feb 22	4.54	25.38	117.07	15.94	27.94	12.0	− 2.06	45.94
Mar 22	4.45	25.3	117.49	15.9	27.99	12.09	− 2.24	46.12
Apr 22	4.51	25.39	116.51	15.93	28.15	12.22	− 2.41	46.48

Presented are the minimum, mean and maximum values as well as the lower and the upper part of the boxplot (as defined in Wikipedia (2023)) Q1 (representing the 25% quartile) and Q3 (representing the 75% quartile). Furthermore, the interquartile range (*IQR*) and the whiskers ($Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$) are given. The whiskers represent the end of the boxplot lines, which mark the border to the data defined as outlier

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Author Contributions GZ: Conceptualization, Validation, Methodology, Writing - Original Draft, Supervision, Project administration. TC: Conceptualization, Validation, Methodology, Formal analysis,

Writing Review. SC: Methodology, Validation, Formal analysis, Writing Review. MF: Methodology, Validation, Formal analysis. JM: Conceptualization, Validation, Writing Review and Editing

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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